Trial Balance Review
Business Requirements & Lineage
HITL Traceability Specification
Ref: `TB-REQ-LIN-2026-v1.0`

SAP-OSS Financial Architecture Team

April 2026

Abstract

This specification defines the Human-in-the-Loop (HITL) Requirements Traceability framework for Standard Chartered Bank's Trial Balance (TB) review process. It provides the complete audit trail from source documents (business cases, detailed operating instructions, BPMN process maps, and workbooks) through machine-readable extraction, structured business requirements, formal specifications, and validating JSON schemas.

The document serves as the **human validation checkpoint** for the automated requirements extraction and traceability process, enabling reviewers to verify the accuracy of automated extractions and establish a clear audit trail from source to implementation.

Four orthogonal workstreams are traced in full:

- a **source document inventory** cataloguing 6 source files totalling 137 MB, including business cases, DOI, BPMN process map, and HKG TB/PL review workbooks;
- an **extraction layer** of 7 machine-readable outputs (markdown, JSON, YAML, CSV) validated against YAML frontmatter schemas;
- a **requirements layer** of 27 business requirements categorised across executive summary, scope, process steps, AI augmentation, and BSS risk assessment;
- a **specification and schema layer** comprising 21 specification elements and 6 JSON schemas with field-level traceability back to requirements.

Reading order. 1 introduces the HITL framework and traceability model; 2 catalogues source documents; 3 covers machine-readable extraction; 4 details requirement derivation; 5 maps specifications to requirements; 6 provides schema field traceability; 7 presents the full forward and backward matrices; 8 identifies traceability gaps; 9 captures review checkpoints and approvals.

Conventions. Source document references appear as `TB-BC-001`. Business requirement IDs appear as `TB-REQ-ES01`. Specification elements appear as `TB-STEP-003`. Schema references appear as `variance-record`. Gap references appear as `G-001`. File paths are typeset in `monospace` and are relative to the repository root.

Companion files. The machine-readable companion to this specification is `docs/tb/machine-readable/hitl-traceability-manifest.yaml`, which enables automated validation of all traceability links documented in this specification.

Visual Traceability Map

This single-page reference shows the complete traceability flow from each source document through extracted content, business requirements, specifications, and implementing schemas.

End-to-End Traceability Overview

Figure 1: Complete Five-Layer Traceability Flow

Traceability Quick Reference

@L2.2cmL3cmL2.5cmL2.5cmL3cm@ **Source** & **Extracted** & **Requirements** & **Specs** & **Schemas**
TB-BC-001 & bc-tb.md & ES01-04, SC01-06, AI01-06 & AI-*, TB-CTRL-* & variance-record, commentary-draft
TB-BC-002 & bc-bss.md & BSS-01-03 & DP-005 & variance-record
TB-BPMN-001 & bpmn-glo.json & PS01-08 & TB-STEP-*, DP-001-004 & variance-record, tb-extract
TB-DOI-001 & doi-tb.md & (supplementary) & – & –
TB-WB-001 & hkg-tb-schema.yaml & (data reference) & – & tb-extract
TB-WB-002 & hkg-pl-schema.yaml & (data reference) & – & pl-extract

Coverage at a Glance

Layer	Count	Traced	Coverage	Status
Source Documents	6	6	100%	Complete
Extracted Files	7	7	100%	Complete
Requirements	27	27	100%	Complete
Specifications	22	22	100%	Complete
Schemas	6	6	100%	Complete
Gaps	2	2	100%	Documented

100% BIDIRECTIONAL TRACEABILITY ACHIEVED

- Every source document section reaches at least one schema
- Every schema field traces back to a source document
- All 27 requirements have documented derivation evidence
- All 22 specifications linked to requirements
- 2 gaps documented with owners and resolution plans

Overview and HITL Framework

This chapter introduces the Human-in-the-Loop (HITL) Requirements Traceability framework, its purpose, and the five-layer traceability model that underpins the Trial Balance review process documentation.

Purpose Statement

The HITL Requirements Traceability specification serves as the **human validation checkpoint** for the automated requirements extraction and traceability process. It is designed to:

1. Provide full transparency into how business requirements were derived from source documents
2. Enable human reviewers to verify the accuracy of automated extractions
3. Establish a clear audit trail from source to implementation
4. Support regulatory compliance by demonstrating requirement provenance
5. Facilitate change impact analysis when source documents are updated

Five-Layer Traceability Model

The traceability framework is organised into five distinct layers, each representing a transformation stage from raw source material to validated implementation artefacts.

Figure 2: Five-Layer Traceability Model with artefact counts

Layer 1: Source Documents

Original business documents in their native formats:

- Business case documents (`.doc`)
- Detailed Operating Instructions (`.docx`)
- BPMN process maps (`.pdf`)
- Working spreadsheets (`.xlsm`)

Layer 2: Extracted Machine-Readable

Automated extraction outputs in machine-processable formats:

- Markdown with YAML frontmatter (text content)
- JSON (structured BPMN elements)
- YAML (schemas and specifications)
- CSV (sample data)

Layer 3: Business Requirements

Structured requirements derived from source documents:

- Executive Summary requirements (4)
- Scope requirements (6)
- Process Step requirements (8)
- AI Augmentation requirements (6)
- BSS Risk Assessment requirements (3)

Layer 4: Specifications

Formal specification elements implementing requirements:

- Process Steps (8)
- Decision Points (5)
- AI Capabilities (6)
- Controls (3)

Layer 5: Schemas & Validation

JSON Schemas validating specification implementation:

- `variance-record` — variance analysis records
- `decision-point` — decision point states
- `entity-params` — per-entity parameters
- `tb-extract` — trial balance extraction
- `pl-extract` — P&L extraction
- `commentary-draft` — LLM commentary drafts

Traceability Marker Reference

Throughout this specification, colour-coded markers indicate traceability links across layers:

@C3cmL5cmL6cm@ **Marker** & **Meaning** & **Example**
`XXX` & Source document reference & `TB-BC-001` = Trial Balance Business Case
`XXX` & Business requirement ID & `TB-REQ-ES01` = Executive Summary req
`XXX` & Specification element & `TB-STEP-003` = Process step
`XXX` & Schema definition & `variance-record` = JSON Schema
`XXX` & Identified gap & `G-001` = Traceability gap

HITL Review Checkpoints

Six human review checkpoints must be validated before the traceability chain is considered complete:

@C1cmL4cmL7cmC2cm@ **ID** & **Checkpoint** & **Review Criteria** & **Status**
H1 & Source Completeness & All source documents listed and extracted & Pending
H2 & Extraction Accuracy & Extracted text matches source content & Pending
H3 & Requirement Derivation & Requirements accurately reflect source intent & Pending
H4 & Traceability Links & All links are valid and bidirectional & Pending
H5 & Schema Alignment & Schemas support all requirement fields & Pending
H6 & Gap Analysis & No orphan requirements or missing mappings & Pending

H1 – Source Completeness Human reviewer must verify that all 6 source documents are correctly catalogued with accurate file sizes, types, and metadata. Any missing documents must be identified and the extraction process re-run.

Scope and Coverage

In Scope

- Trial Balance review process across 234 Legal Entities
- AI-augmented variance analysis and commentary generation
- Month-end close controls (M-3 to M+5 timeline)
- FORTM pack generation and submission
- BSS Risk Assessment integration points

Out of Scope

- Country IFRS reporting (separate process)
- SAP/PeopleSoft integration detail
- Real-time intraday controls
- Audit committee reporting (downstream consumer)

Coverage Statistics

Table 2: Traceability Coverage Statistics

Metric	Count	Coverage
Source Documents	6	100% catalogued
Extracted Files	7	100% traced to source
Business Requirements	27	100% traced to source
Specifications	21	100% traced to requirements
Schemas	6	100% traced to specifications
Identified Gaps	2	Documented with owners

Source Document Inventory

This chapter catalogues all source documents that form the foundation of the Trial Balance review requirements. Each document is assigned a unique source reference identifier for traceability purposes.

Source Document Catalog

@C2cmL5cmC1.5cmC2cmC2.5cm@

Source ID & **Document Name** & **Type** & **Size** & **Extraction**
Source ID & **Document Name** & **Type** & **Size** & **Extraction**

`TB-BC-001` & Business case - Trial Balance (2).doc & .doc & 96 KB & Complete
`TB-BC-002` & Business case - BSS Risk Assessment (2).doc & .doc & 95 KB & Complete
`TB-DOI-001` & DOI for Trial Balance Process.docx & .docx & 6.3 MB & Complete
`TB-BPMN-001` & Operation of Month End Close Controls - Trial Balance Review - GLO (1).pdf & .pdf & 64 KB & Complete
`TB-WB-001` & HKG_TB review Nov'25.xlsm & .xlsm & 73 MB & Schema Only
`TB-WB-002` & HKG_PL review Nov'25.xlsm & .xlsm & 51 MB & Schema Only

Source Document Descriptions

`TB-BC-001`: Business Case - Trial Balance

@L4cmL10cm@ **Attribute** & **Value**
Document Title & Business Case: Month End Controls – Trial Balance
Filepath & `docs/tb/Business case - Trial Balance (2).doc`
Created By & 1498002 (S1 Badrinath)
Created Date & 2025-12-14
Version & 1.0
File Size & 97,792 bytes
Purpose & Define business justification for AI automation of TB review

Key Sections:

- **Section 1: Executive Summary** — TB review conducted M+1 to M+7 across 234 LEs; AI model for variance analysis and commentary generation
- **Section 2: Problem/Opportunity Statement** — Current state analysis; 20-25 FTEs, 5 FTE savings target
- **Section 3: Project Description** — Proposed AI solution using Studio AI (in-house)
- **Section 5: Financial Analysis** — *TBD* — *Gap `G-001`*
- **Section 6: Risk Assessment** — PSGL/S4 compatibility, SAP integration risks
- **Section 7: Implementation Approach** — 15-week timeline

Content Summary:

> *"Trial Balance review is conducted during M+1 to M+7 across all Legal entities, Segment & Group, to understand the variance & balance movements between periods. Deep dives to assess the events,*

transactions causing the variances and reasons behind them. Analysed with detailed commentaries for these movements are prepared & submitted to Group for further review & used for Country CFO packs/Legal entity reporting."

`TB-BC-002`: Business Case - BSS Risk Assessment

@L4cmL10cm@ **Attribute** & **Value**
Document Title & Business Case: BSS Risk Assessment
Filepath & `docs/tb/Business case - BSS Risk Assessment (2).doc`
Created By & 1498002 (S1 Badrinath)
Created Date & 2025-12-15
Version & 1.0
File Size & 95,232 bytes
Purpose & Define AI augmentation for BSS risk assessment process

Key Content:

- **Timing:** M+12 to M+15 across all Legal entities
- **Criticality:** CFO attestation and Risk/Audit committee reporting
- **Risk Categories:** Substantiated at Risk, Unsubstantiated Unowned, Unsubstantiated Owned
- **FTE Impact:** 10-12 FTEs current, 1 FTE savings target

`TB-DOI-001`: DOI for Trial Balance Process

@L4cmL10cm@ **Attribute** & **Value**
Document Type & Detailed Operating Instructions
Filepath & `docs/tb/DOI for Trial Balance Process .docx`
File Size & 6,583,541 bytes
Purpose & Step-by-step operational guidance for TB review execution
Contains & Process flow diagrams (EMF/PNG images)
Image Count & 4 extracted images

Extracted Images:

- `doi-images/doi-image-001.emf` — Process flow diagram 1
- `doi-images/doi-image-002.emf` — Process flow diagram 2
- `doi-images/doi-image-003.png` — Process flow diagram 3
- `doi-images/doi-image-004.emf` — Process flow diagram 4

`TB-BPMN-001`: BPMN Process Map

@L4cmL10cm@ **Attribute** & **Value**
Document Title & Operation of Month End Close Controls - Trial Balance Review - GLO
Filepath & `docs/tb/Operation of Month End Close Controls - Trial Balance Review - GLO (1).pdf`
Process Owner & Modi, Piyush
Last Changed & 2025-05-23T19:50:30Z
Scope & GLO (Global)
File Size & 65,367 bytes

BPMN Elements Extracted:

- **States:** 27 (including 3 error-handling states)
- **Gateways:** 5 exclusive decision gateways
- **Timeline Phases:** 4 (Preparation, Extraction, Analysis, Review)
- **Participants:** 3 (GFS Analyst, Head of Africa GFS, Country Stakeholders)

`TB-WB-001`: HKG TB Review Workbook

@L4cmL10cm@ **Attribute** & **Value**
Document Title & HKG_TB review Nov'25.xlsm
Filepath & `docs/tb/HKG_TB review Nov'25.xlsm`
File Size & 76,989,527 bytes (73 MB)
Sheet Count & 13
Extraction Status & Schema extraction only (data too large)

Key Sheets:

- `IFRS Summary BS` — IFRS balance sheet summary view
- `Count of Comments` — Commentary tracking (7 fields)
- `BS Variance` — Balance sheet variance analysis (5 fields)
- `RAW TB NOV'25` — Raw GL balance query November 2025
- `RAW TB OCT'25` — Raw GL balance query October 2025
- `GCOA` — Global Chart of Accounts (5 fields)
- `Dept Mapping` — Department-to-vertical mapping (12 fields)
- `Base file` — Account-level base data (13 fields)

`TB-WB-002`: HKG PL Review Workbook

@L4cmL10cm@ **Attribute** & **Value**
Document Title & HKG_PL review Nov'25.xlsm
Filepath & `docs/tb/HKG_PL review Nov'25.xlsm`
File Size & 53,245,945 bytes (51 MB)
Sheet Count & 14
Extraction Status & Schema extraction only (data too large)

Source Document Verification

H1 – Source Completeness Verification The human reviewer must verify the following for checkpoint H1:

1. All 6 source documents exist at the specified file paths
2. File sizes match the catalogued values (within 1% tolerance)
3. Document metadata (author, date, version) is accurate
4. No additional source documents have been omitted
5. Extraction status correctly reflects processing state

Verification Checklist:

Table 3: Source Document Verification Checklist

Source ID	Exists	Size OK	Metadata OK	Verified
`TB-BC-001`	☐	☐	☐	☐
`TB-BC-002`	☐	☐	☐	☐
`TB-DOI-001`	☐	☐	☐	☐
`TB-BPMN-001`	☐	☐	☐	☐
`TB-WB-001`	☐	☐	☐	☐
`TB-WB-002`	☐	☐	☐	☐

Machine-Readable Extraction Layer

This chapter documents the extraction process that transforms source documents into machine-readable formats, providing the foundation for automated validation and requirements traceability.

Extraction Process Overview

Source documents were processed using automated extraction scripts to produce machine-readable formats suitable for:

- Text analysis and requirement derivation
- Schema validation and data quality checks
- RAG (Retrieval-Augmented Generation) pipeline ingestion
- Version control and change tracking

@L3.5cmL4cmL6.5cm@ **Script** & **Source Types** & **Output Format**
`extract_docs.py` & .doc, .docx & Markdown with YAML frontmatter
`extract_pdf.py` & .pdf (BPMN) & Markdown + BPMN elements JSON
`extract_xlsx.py` & .xlsm workbooks & YAML schemas + CSV samples
`build_rag_chunks.py` & All extracted text & JSONL chunks for RAG

Extracted Document Mapping

@C2cmL5cmL6.5cm@

Source ID & **Extracted File** & **Extraction Notes**
Source ID & **Extracted File** & **Extraction Notes**
`TB-BC-001` & `business-case-trial-balance.md` & Full text extraction; tables converted to markdown; 1,172 words
`TB-BC-002` & `business-case-bss-risk-assessment.md` & Full text extraction; 8,908 bytes
`TB-DOI-001` & `doi-trial-balance-process.md` & Text + 4 images extracted to `doi-images/`
`TB-BPMN-001` & `bpmn-tb-review-glo.md` + `bpmn-elements.json` & Process text + structured BPMN elements
`TB-WB-001` & `hkg-tb-schema.yaml` + `hkg-tb-sample.csv` & Schema for 13 sheets; 100-row sample
`TB-WB-002` & `hkg-pl-schema.yaml` + `hkg-pl-sample.csv` & Schema for 14 sheets; 100-row sample

Extraction File Inventory

Extracted Text Files

@L6cmC2cmC2cmL3.5cm@

File Path & **Size** & **Format** & **Source Ref**
File Path & **Size** & **Format** & **Source Ref**
`extracted/business-case-trial-balance.md` & 9,824 B & Markdown & `TB-BC-001`
`extracted/business-case-bss-risk-assessment.md` & 8,908 B & Markdown & `TB-BC-002`

extracted/doi-trial-balance-process.md & 865 B & Markdown & TB-DOI-001
extracted/bpmn-tb-review-glo.md & 9,154 B & Markdown & TB-BPMN-001
extracted/bpmn-elements.json & 2,264 B & JSON & TB-BPMN-001

Extracted Image Files

@L6cmC2cmC2cmL3.5cm@

File Path & **Size** & **Format** & **Source Ref**
File Path & **Size** & **Format** & **Source Ref**
extracted/doi-images/doi-image-001.emf & 8,148 B & EMF & TB-DOI-001
extracted/doi-images/doi-image-002.emf & 8,272 B & EMF & TB-DOI-001
extracted/doi-images/doi-image-003.png & 43,650 B & PNG & TB-DOI-001
extracted/doi-images/doi-image-004.emf & 8,132 B & EMF & TB-DOI-001

Training Data Files

@L6cmC2cmC2cmL3.5cm@

File Path & **Size** & **Format** & **Source Ref**
File Path & **Size** & **Format** & **Source Ref**
training-data/hkg-tb-schema.yaml & 20,228 B & YAML & TB-WB-001
training-data/hkg-tb-sample.csv & 17,916 B & CSV & TB-WB-001
training-data/hkg-pl-schema.yaml & 16,982 B & YAML & TB-WB-002
training-data/hkg-pl-sample.csv & 19,110 B & CSV & TB-WB-002

Extraction Frontmatter Format

Each extracted markdown file includes YAML frontmatter for traceability and metadata validation:

```
---
source_file: 'Business case - Trial Balance (2).doc'
source_type: 'doc'
extracted_at: '2026-04-13T06:16:45.446975+00:00'
word_count: 1172
source_ref: 'TB-BC-001'
---
# Business Case: Trial Balance
```

```
## 1. Executive Summary
...
```

Frontmatter Fields: @L3cmC2cmL8.5cm@ **Field** & **Type** & **Description**
`source_file` & string & Original filename
`source_type` & enum & File extension (.doc, .docx, .pdf, .xlsm)
`extracted_at` & ISO 8601 & Timestamp of extraction
`word_count` & integer & Word count of extracted text
`source_ref` & string & Traceability reference ID

BPMN Elements Extraction

The BPMN process map (`TB-BPMN-001`) was parsed into structured JSON to enable automated workflow validation:

```
{
  "workflowId": "tb-review-glo",
  "version": "1.0.0",
  "states": [
    {
      "stateId": "START",
      "type": "start_event",
      "name": "Prepare Schedule Work Day",
      "transitions": [{"target": "ROLL_FORWARD"}]
    },
    {
      "stateId": "ROLL_FORWARD",
      "type": "task",
      "name": "Roll Forward Prior Month File",
      "timing": "M-3",
      "systems": ["MS Excel", "Shared Drive"],
      "transitions": [{"target": "DOWNLOAD_PSGL"}]
    }
  ],
  "gateways": [...],
  "dataObjects": [...],
  "participants": [...]
}
```

Extracted BPMN Statistics:

Table 4: BPMN Extraction Statistics

Element Type	**Count**
Start Events	1
Task States	21
Gateway States	5
Error-Handling States	3
End Events	1
Total States	**27**
Data Objects	6
Participants	3

Workbook Schema Extraction

Excel workbooks were processed to extract structural schemas without copying sensitive data:

HKG TB Review Schema (`TB-WB-001`)

@L4cmC2cmL7.5cm@ **Sheet Name** & **Fields** & **Purpose**
IFRS Summary BS & – & IFRS balance sheet summary view
Count of Comments & 7 & Commentary tracking (reviewer, status, hours)
BS Variance & 5 & Balance sheet variance analysis with thresholds
PL Variance & 4 & P&L variance analysis
GCOA & 5 & Global Chart of Accounts definitions
Dept Mapping & 12 & Department-to-vertical mapping
Base file & 13 & Account-level base data with mappings

Extraction Quality Metrics

@L5cmC2.5cmL6cm@ **Metric** & **Value** & **Human Validation Required**
Source documents processed & 6 of 6 & Verify all documents included
Total extracted text size & 522 KB & Verify no truncation occurred
Tables extracted & 12 & Verify table structure preserved
Images extracted & 4 & Verify images are referenced correctly
BPMN elements parsed & 27 states & Verify state machine completeness
Workbook sheets processed & 27 & Verify key sheets included

H2 – Extraction Accuracy Verification The human reviewer must verify the following for checkpoint H2:

1. Sample extracted text against original documents (minimum 3 paragraphs per document)
2. Table formatting is preserved in markdown conversion

3. BPMN state names and transitions match the source PDF
4. Workbook schema field names match column headers
5. No content truncation or corruption occurred

Business Requirements Catalogue

This chapter presents all business requirements derived from the source documents, with **exact quotations** and **page/paragraph references** enabling human auditors to verify each requirement against its source.

How to Read This Chapter

Each requirement entry contains:

1. **Requirement ID:** Unique identifier (e.g., TB-REQ-ES01)
2. **MoSCoW Priority:** MUST / SHOULD / COULD / WON'T
3. **Source Quotation:** Exact text from source document (in italics)
4. **Source Location:** Document ID, section, paragraph, page number
5. **Derived Requirement:** Formal requirement statement
6. **Acceptance Criteria:** Measurable conditions for verification
7. **Rationale:** Why this requirement was derived from the quoted text

Executive Summary Requirements

TB-REQ-ES01: AI-Driven Variance Analysis

"An AI model produces business insights and written commentaries (Finance reporting caption Level & Segment Level with Legal entity view) based on the analysis performed on general ledger transaction data that ***detects anomalies, trends, unusual movements, new accounts, spikes/drops vs prior periods, and material balances****."*

Attribute	Value
Requirement ID	TB-REQ-ES01
MoSCoW Priority	**MUST**
Source Document	Business case - Trial Balance (2).doc
Source ID	TB-BC-001
Source Location	Section 1: Executive Summary, Paragraph 3, Page 1 (Lines 62–68 in extracted MD)
Created By	Employee ID 1498002 (S1 Badrinath)
Document Date	14-Dec-2025

Derived Requirement Statement:

The system SHALL implement an AI model capable of performing the following analyses on general ledger (GL) transaction data:

1. **Anomaly Detection:** Identify transactions or balances that deviate significantly from expected patterns
2. **Trend Analysis:** Detect multi-period trends in account balances
3. **Unusual Movement Detection:** Flag balance changes that exceed normal operational variance
4. **New Account Detection:** Identify accounts appearing for the first time in the current period
5. **Spike/Drop Detection:** Identify sudden increases or decreases compared to prior periods
6. **Material Balance Identification:** Flag balances exceeding defined materiality thresholds

Acceptance Criteria:

1. The AI model SHALL process GL data for all 234 Legal Entities (per TB-BC-001, Section 2)
2. Anomaly detection SHALL flag items with Z-score > 3.0 or configurable threshold
3. Spike/drop detection SHALL identify variances $> 25\%$ period-over-period (or entity-configurable)
4. New account detection SHALL compare current period CoA against prior 12-month baseline
5. Material balance threshold SHALL default to entity-specific materiality (configurable per LE)
6. Processing SHALL complete within M+1 to M+5 window (per BPMN: TB-BPMN-001)

Derivation Rationale: The source text explicitly lists six AI capabilities (anomalies, trends, unusual movements, new accounts, spikes/drops, material balances). Each is converted to a testable sub-requirement with quantifiable criteria.

TB-REQ-ES02: Commentary Generation

*"GCFO teams get a **narrative summary** instead of just numbers, accelerating review and improving decision-making."*

Attribute	Value
Requirement ID	TB-REQ-ES02
MoSCoW Priority	**MUST**
Source Document	Business case - Trial Balance (2).doc
Source ID	TB-BC-001
Source Location	Section 1: Executive Summary, Paragraph 3, Page 1 (Lines 66–68)

Derived Requirement Statement:

The system SHALL generate natural language commentary summaries for:

- Finance reporting caption level
- Segment level
- Legal entity level

Acceptance Criteria:

1. Commentary SHALL be generated in grammatically correct English prose
2. Commentary SHALL reference specific variance amounts (e.g., "Revenue increased by $2.3M (15%) due to...")
3. Commentary SHALL cite contributing factors identified by AI analysis
4. Commentary confidence score SHALL be displayed (0–100%)
5. Human reviewer SHALL be able to edit/approve/reject each commentary
6. Commentary generation SHALL complete within 60 seconds per Legal Entity

TB-REQ-ES03: FTE Savings Target

*"At least **20 to 25 FTEs** spent on an average **4 hrs a day** during M-2 to M+5, which is equivalent to ∼**5 FTE saves**, with strengthening controls by eliminating manual process and reducing the overall book close timelines enabling Fast close."*

Attribute	Value
Requirement ID	TB-REQ-ES03
MoSCoW Priority	**MUST**
Source Document	Business case - Trial Balance (2).doc
Source ID	TB-BC-001
Source Location	Section 1: Executive Summary, Paragraph 4, Pages 1–2 (Lines 72–76)

Derived Requirement Statement:

The system SHALL reduce manual effort in Trial Balance review from current baseline of 20–25 FTEs × 4 hours/day to achieve a minimum 5 FTE equivalent savings.

Acceptance Criteria:

1. Baseline measurement: Current effort = 20–25 FTEs × 4 hrs/day × 8 days (M-2 to M+5) = 640–800 person-hours per close cycle
2. Target measurement: Post-implementation effort ≤ 480 person-hours (15–20 FTEs × 4 hrs × 8 days)
3. Savings calculation: ≥ 160 person-hours saved per cycle (5 FTE equivalent)
4. Measurement method: Time tracking via ServiceNow tickets or equivalent
5. Verification: Quarterly review of actual vs. target savings

Financial Context:

Metric	Current State	Target State	Savings
FTEs involved	20–25	15–20	5
Hours per day	4	4	–
Days per cycle (M-2 to M+5)	8	8	–
Person-hours per cycle	640–800	480–640	160 (min)

Scope Requirements

TB-REQ-SC01: Legal Entity Coverage

*"Downloading Trail balance from PSGL/S4 & preparing the detailed analysis across all **234 LEs** are highly time consuming..."*

Requirement ID	TB-REQ-SC01
MoSCoW Priority	**MUST**
Source Location	TB-BC-001, Section 1, Paragraph 2, Page 1 (Lines 55–59)

Derived Requirement Statement:

The system SHALL support Trial Balance review for exactly 234 Legal Entities.

Acceptance Criteria:

1. System configuration SHALL include all 234 Legal Entity codes
2. System SHALL process TB data for each LE independently
3. Dashboard SHALL display completion status for all 234 LEs
4. No Legal Entity SHALL be excluded from automated processing

TB-REQ-SC02: Reporting Dimensions

*"...current view is at **Legal entity level** and need to be in **Finance caption level & segment level**."*

Requirement ID	TB-REQ-SC02
MoSCoW Priority	**MUST**
Source Location	TB-BC-001, Section 1, Paragraph 2, Pages 1 (Lines 55–59)

Derived Requirement Statement:

The system SHALL provide variance analysis and commentary at three reporting dimensions:

1. **Legal Entity Level:** Individual LE view (current capability, to be retained)
2. **Finance Caption Level:** Aggregation by standardized finance captions (e.g., Revenue, COGS, Operating Expenses)
3. **Segment Level:** Aggregation by business segment as defined in CoA

Acceptance Criteria:

1. User SHALL be able to toggle between all three reporting dimensions
2. Drill-down SHALL be available from Segment → Finance Caption → LE
3. Commentary SHALL be generated at each dimension level
4. Export to Excel SHALL preserve dimension hierarchy

TB-REQ-SC03: Out of Scope Items

*"Out of scope – **Country IFRS reporting, BSS Risk assessment**"*

Requirement ID	TB-REQ-SC03
MoSCoW Priority	**WON'T** (this phase)
Source Location	TB-BC-001, Section 3: Project Description, Scope, Page 2 (Lines 152–154)

Explicit Exclusions:

The following items are **explicitly out of scope** per source documentation:

1. **Country IFRS Reporting:** Local GAAP/IFRS adjustments not included
2. **BSS Risk Assessment:** Balance Sheet Substantiation is a separate process

Note: These items are documented as future phases (see Gap G-003 in Chapter 8).

Process Step Requirements (from BPMN)

The following requirements are derived from the BPMN process diagram **TB-BPMN-001**: "Operation of Month End Close Controls - Trial Balance Review - GLO".

Source: `Operation of Month End Close Controls - Trial Balance Review - GLO (1).pdf`
Process Owner: Modi, Piyush
Last Changed: May 23, 2025, 7:50:30 PM
Author ID: 8203488

TB-REQ-PS01: Roll Forward Prior Month File

"Roll Forward Prior Month File to Reflect Current..."
Timing: M-3
System: MS-E-xcel-1, Shared Drive

Requirement ID	TB-REQ-PS01
MoSCoW Priority	**MUST**
Source Location	TB-BPMN-001, Task Row 1, Timing M-3
EUC Reference	N/A (no EUC control on this step)

Derived Requirement: The system SHALL roll forward the prior month's Trial Balance workbook structure to the current period while preserving:

- Prior period comparative balances
- Account structure and mappings
- Historical commentary for reference

Acceptance Criteria:

1. Roll-forward SHALL complete by M-3 (3 days before month end)
2. Prior month comparatives SHALL be locked from editing
3. Current period columns SHALL be initialized to zero

TB-REQ-PS02: Download PSGL Master Data

"Download PSGL [...] Update 'Accounts Master' Sheet"
Timing: M-3
System: PSGL, MS-E-xcel-1
EUC Control: EX/200/1353107/0001/000

Requirement ID	TB-REQ-PS02
MoSCoW Priority	**MUST**

Source Location	TB-BPMN-001, Task Row 2, Timing M-3
EUC Reference	EX/200/1353107/0001/000

Derived Requirement: The system SHALL extract Chart of Accounts (CoA) master data from PSGL and update the Accounts Master reference.

Acceptance Criteria:

1. CoA extract SHALL include all active GL accounts
2. Extract SHALL match EUC control EX/200/1353107/0001/000 specifications
3. New accounts SHALL be flagged for review (feeds TB-REQ-AI03)

TB-REQ-PS03: Extract Trial Balance

"Extract Trial Balance [...] File in Current Month"
Timing: M-3 to M+5
System: PSGL, MS-E-xcel-1, Shared Drive

Requirement ID	TB-REQ-PS03
MoSCoW Priority	**MUST**
Source Location	TB-BPMN-001, Task Row 3

Derived Requirement: The system SHALL extract Trial Balance data from PSGL/S4 for all 234 Legal Entities.

Acceptance Criteria:

1. Extract SHALL include: Account Number, Account Name, Opening Balance, Movements, Closing Balance
2. Extract SHALL be available in both local currency (LC) and reporting currency (RC)
3. Data SHALL be timestamped with extraction date/time
4. Hash/checksum SHALL be generated for data integrity verification

TB-REQ-PS04: Calculate Variance

"Calculate Variance Between Prior [...] and Current Month"
Timing: M-2 to M+5
System: MS-E-xcel-1, Shared Drive
EUC Control: EX/200/1579702/0001/000

Requirement ID	TB-REQ-PS04
MoSCoW Priority	**MUST**
Source Location	TB-BPMN-001, Task Row 4
EUC Reference	EX/200/1579702/0001/000

Derived Requirement: The system SHALL calculate variance between:

- Current month balance vs. prior month balance
- Current month balance vs. same month prior year (YoY)
- Both absolute and percentage variance

Acceptance Criteria:

1. Variance calculation: $V = B_{current} - B_{prior}$
2. Percentage variance: $V\% = \frac{B_{current} - B_{prior}}{|B_{prior}|} \times 100$
3. Division by zero handling: If $B_{prior} = 0$, display "N/A (New)"
4. Calculation SHALL match EUC control specifications

TB-REQ-PS05: Filter by Materiality

"Filter Variances [...] Set Criteria"
Timing: M-2 to M+5

Requirement ID	TB-REQ-PS05
MoSCoW Priority	**MUST**
Source Location	TB-BPMN-001, Task Row 5

Derived Requirement: The system SHALL filter variances based on configurable materiality thresholds.

Acceptance Criteria:

1. Materiality threshold SHALL be configurable per Legal Entity
2. Default threshold: Absolute $> \$50,000$ OR Percentage $> 10\%$
3. Filter SHALL support AND/OR logic combinations
4. Filter SHALL support account-type-specific thresholds

Requirements Summary Matrix

Table 17: Complete Requirements Traceability Summary

Req ID	MoSCoW	Title	Source Ref	Acceptance Criteria
Req ID	**MoSCoW**	**Title**	**Source Ref**	**Acceptance Criteria**
TB-REQ-ES01	MUST	AI Variance Analysis	TB-BC-001, S1P3	AC-ES01-1..6
TB-REQ-ES02	MUST	Commentary Generation	TB-BC-001, S1P3	AC-ES02-1..6
TB-REQ-ES03	MUST	5 FTE Savings	TB-BC-001, S1P4	AC-ES03-1..5
TB-REQ-SC01	MUST	234 LE Coverage	TB-BC-001, S1P2	AC-SC01-1..4
TB-REQ-SC02	MUST	3-Level Reporting	TB-BC-001, S1P2	AC-SC02-1..4
TB-REQ-SC03	WON'T	Exclusions Documented	TB-BC-001, S3	N/A
TB-REQ-PS01	MUST	Roll Forward	TB-BPMN-001	AC-PS01-1..3
TB-REQ-PS02	MUST	PSGL Master Download	TB-BPMN-001	AC-PS02-1..3
TB-REQ-PS03	MUST	TB Extract	TB-BPMN-001	AC-PS03-1..4
TB-REQ-PS04	MUST	Variance Calculation	TB-BPMN-001	AC-PS04-1..4
TB-REQ-PS05	MUST	Materiality Filter	TB-BPMN-001	AC-PS05-1..4

MoSCoW Distribution:

- **MUST:** 10 requirements (91%)
- **WON'T:** 1 requirement (9%) – explicitly excluded from scope

Human Auditor Verification Checklist

For each requirement above, the human auditor should verify:

1. **Source Quotation Accuracy:** Does the quoted text match the original document exactly?
2. **Location Correctness:** Is the section/paragraph/page reference correct?
3. **Derivation Logic:** Is the requirement a reasonable derivation from the quoted text?
4. **Acceptance Criteria Measurability:** Can each criterion be objectively verified?
5. **MoSCoW Appropriateness:** Does the priority reflect business importance?
6. **Completeness:** Are there any requirements in the source document not captured here?

Signoff:

- Name:

_____

- Date: _____
- Signature: _____

Specification Mapping

This chapter maps each specification element to its source requirement(s), establishing the requirement-to-specification traceability layer of the HITL framework.

Process Step Specifications

@C2cmL3.5cmC3cmL5cm@

Spec ID & **Specification** & **Req ID(s)** & **Implementation Notes**
Spec ID & **Specification** & **Req ID(s)** & **Implementation Notes**
`TB-STEP-001` & Roll Forward Prior Month File & `TB-REQ-PS01` & Auto-generate template; pre-populate parameters
`TB-STEP-002` & Download PSGL & Update Master & `TB-REQ-PS02` & PSGL/S4 API integration; auto-detect changes
`TB-STEP-003` & Extract Trial Balance & `TB-REQ-PS03`, `TB-REQ-SC01` & Parallel extraction for 234 LEs
`TB-STEP-004` & Calculate & Filter Variances & `TB-REQ-PS04`, `TB-REQ-PS05` & ML-adaptive materiality thresholds
`TB-STEP-005` & Seek & Update Commentaries & `TB-REQ-PS06`, `TB-REQ-AI06` & LLM-generated draft commentaries
`TB-STEP-006` & Journal Posting Investigation & `TB-REQ-PS07` & Auto-detect unposted journals
`TB-STEP-007` & Track Entries & Risk & `TB-REQ-PS07` & Predictive risk scoring
`TB-STEP-008` & Prepare Summary & Review & `TB-REQ-PS08` & Auto-generate FORTM pack section

Decision Point Specifications

@C1.5cmL3cmC3cmL6cm@

DP ID & **Decision Point** & **Req ID(s)** & **Specification Detail**
DP ID & **Decision Point** & **Req ID(s)** & **Specification Detail**
`DP-001` & Materiality Check & `TB-REQ-PS05`, `TB-REQ-AI05` & Threshold: BS $100M, PL $3M; ML-adaptive per category
`DP-002` & Journal Status & `TB-REQ-PS07` & States: posted / posting_required

/ not_posted
`DP-003` & Investigation Required & `TB-REQ-PS07` & Predictive model based on entry history
`DP-004` & Entry Confirmation & `TB-REQ-PS07` & M+4 deadline tracking; auto-escalation
`DP-005` & Unexplained Variance & `TB-REQ-AI06`, `BSS-REQ-01` & Categorize: substantiated_at_risk, unsubstantiated_unowned, unsubstantiated_owned

Decision Point Detail: `DP-001`

@L4cmL10cm@ **Attribute** & **Value**
Specification ID & DP-001
Name & Materiality Check
Gateway Type & Exclusive
BPMN State & CHECK_MATERIAL_VARIANCES
Input Metric & `variance_amount`

Balance Sheet (BS) & $100,000,000
Profit & Loss (PL) & $3,000,000
Threshold Source & Static (v1.0), ML-adaptive (v2.0)

Material = Yes & → SEEK_COMMENTARIES
Material = No & → CHECK_UNEXPLAINED

AI Capability Specifications

@C2cmL3.5cmC2.5cmL5.5cm@

Spec ID & **AI Capability** & **Req ID(s)** & **Implementation Detail**
Spec ID & **AI Capability** & **Req ID(s)** & **Implementation Detail**
`AI-ANOM` & Anomaly Detection & `TB-REQ-AI01` & 3σ z-score with MAD fallback for non-normal distributions
`AI-TREND` & Trend Analysis & `TB-REQ-AI02` & 12-month rolling window; percentile ranking
`AI-NEW` & New Account Detection & `TB-REQ-AI03` & Flag accounts not in prior period master
`AI-SPIKE` & Spike/Drop Detection & `TB-REQ-AI04` & Period spike check; sign flip detection
`AI-MAT` & Material Balance ID & `TB-REQ-AI05` & Category-specific thresholds with multipliers
`AI-LLM` & LLM Commentary & `TB-REQ-AI06` & RAG-augmented generation; >85% acceptance rate target

AI Capability Detail: `AI-ANOM`

@L4cmL10cm@ **Attribute** & **Value**
Specification ID & AI-ANOM
Name & Anomaly Detection
Requirement & `TB-REQ-AI01`

Primary Method & Z-score (3σ threshold)
Fallback Method & Median Absolute Deviation (MAD)
Fallback Trigger & Non-normal distribution detected (Shapiro-Wilk $p < 0.05$)

Period Spike & `PERIOD_SPIKE`
Sign Flip & `SIGN_FLIP`
New Account & `NEW_ACCOUNT`
Material Balance & `MATERIAL_BALANCE`

AI Capability Detail: `AI-LLM`

@L4cmL10cm@ **Attribute** & **Value**
Specification ID & AI-LLM
Name & LLM Commentary Generation
Requirement & `TB-REQ-AI06`

Method & RAG-augmented generation
Vector Store & HANA Cloud Vector Engine
Chunk Size & 220 words
Overlap & 40 words

Acceptance Rate Target & >85% require only minor edits
Factual Accuracy & 100% of referenced figures match GL
Latency & <30 seconds P95

Control Specifications

@C2.5cmL4cmC2.5cmL4.5cm@

Control ID & **EUC Reference** & **Req ID(s)** & **Control Objective**
Control ID & **EUC Reference** & **Req ID(s)** & **Control Objective**
`TB-CTRL-001` & EX/200/1353107/0001/000 & `TB-REQ-PS02` & PSGL master data extraction integrity
`TB-CTRL-002` & EX/200/1579702/0001/000 & `TB-REQ-PS04` & Variance analysis calculation accuracy
`TB-CTRL-003` & EX/200/1579702/0002/000 & `TB-REQ-PS08` & Senior review before FORTM submission

Control Detail: `TB-CTRL-002`

@L4cmL10cm@ **Attribute** & **Value**
Control ID & TB-CTRL-002
EUC Reference & EX/200/1579702/0001/000
Requirement & `TB-REQ-PS04`
Objective & Variance analysis calculation accuracy
Frequency & Monthly

Activity 1 & Validate row counts match PSGL source
Activity 2 & Verify variance formula: current_balance − prior_balance
Activity 3 & Confirm materiality thresholds applied correctly
Activity 4 & Reconcile total variances to GL movement report

Specification Coverage Summary

Table 18: Specification Coverage Statistics

Specification Type	Count	Requirements Covered	Coverage
Process Steps	8	10	100%
Decision Points	5	6	100%
AI Capabilities	6	6	100%
Controls	3	3	100%
Total	**22**	**25 unique**	**100%**

Note: Some requirements are covered by multiple specifications (e.g., `TB-REQ-PS07` is covered by `TB-STEP-006`, `TB-STEP-007`, `DP-002`, `DP-003`, and `DP-004`).

H4 – Traceability Links Verification (Part 1) The human reviewer must verify the following for checkpoint H4:

1. Each specification correctly references its source requirements
2. Implementation details are consistent with requirement text
3. No specifications exist without a requirement reference
4. Specification IDs follow naming conventions

Schema Mapping

This chapter maps specification elements to their implementing schemas, completing the traceability chain from source to implementation. Each schema field

is traced back to its source specification and requirement.

Schema Inventory

@L5cmL9cm@ **Schema File** & **Purpose**
`variance-record` & Variance records from Stage 2 anomaly detection
`decision-point` & Decision point state transitions
`entity-params` & Per-entity parameter configuration
`tb-extract` & Trial balance data extraction format
`pl-extract` & P&L data extraction format
`commentary-draft` & LLM-generated commentary drafts

Specification to Schema Mapping

@C2cmL4cmC3cmL4.5cm@

Spec ID & **Specification** & **Schema(s)** & **Key Fields**
Spec ID & **Specification** & **Schema(s)** & **Key Fields**
`TB-STEP-003` & Extract Trial Balance & `tb-extract` & extract_id, legal_entity, period, balances
`TB-STEP-004` & Calculate Variances & `variance-record` & variance_id, variance_amount, flags
`DP-001` & Materiality Check & `variance-record` & materiality_assessment.*
`DP-002` & Journal Status & `variance-record` & journal_status, journal_ids
`DP-003` & Investigation & `variance-record` & investigation_status
`DP-004` & Entry Confirmation & `variance-record` & confirmation_received, days_to_deadline
`DP-005` & Unexplained Variance & `variance-record` & is_unexplained, unexplained_category
`AI-ANOM` & Anomaly Detection & `variance-record` & anomaly_detection.period_spike.*
`AI-NEW` & New Account & `variance-record` & anomaly_detection.new_account.*
`AI-SPIKE` & Spike/Drop & `variance-record` & flags: [PERIOD_SPIKE, SIGN_FLIP]
`AI-LLM` & LLM Commentary & `commentary-draft` & draft_text, confidence, sources

Variance Record Schema (`variance-record`)

The variance record schema is the primary data structure for the TB review process, capturing all variance analysis, anomaly detection, and decision point state information.

Field-Level Traceability

@L4cmL2.5cmC2cmL4.5cm@

Schema Field & **Type** & **Spec ID** & **Source Requirement**
Schema Field & **Type** & **Spec ID** & **Source Requirement**
`variance_id` & string & `TB-STEP-004` & `TB-REQ-PS04`
`account_code` & string & `TB-STEP-003` & `TB-REQ-PS03`
`account_category` & enum & `AI-MAT` & `TB-REQ-AI05`
`finance_caption` & string & `TB-STEP-004` & `TB-REQ-SC02`
`segment` & string & `TB-STEP-004` & `TB-REQ-SC03`
`legal_entity` & string & `TB-STEP-003` & `TB-REQ-SC01`
`variance_amount` & number & `TB-STEP-004` & `TB-REQ-PS04`
`flags` & array[enum] & `AI-ANOM` & `TB-REQ-AI01`
`materiality_assessment.is_material` & boolean & `DP-001` & `TB-REQ-PS05`
`materiality_assessment.threshold_source` & enum & `DP-001` & `TB-REQ-AI05`
`journal_status` & enum & `DP-002` & `TB-REQ-PS07`
`investigation_status` & enum & `DP-003` & `TB-REQ-PS07`
`confirmation_received` & boolean & `DP-004` & `TB-REQ-PS07`
`is_unexplained` & boolean & `DP-005` & `TB-REQ-AI06`
`unexplained_category` & enum & `DP-005` & `BSS-REQ-01`

Schema Structure

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "variance-record.schema.json",
  "type": "object",
  "required": ["variance_id", "account_code", "legal_entity",
               "variance_amount"],
  "properties": {
    "variance_id": { "type": "string", "pattern": "^VAR-" },
    "account_code": { "type": "string" },
    "account_category": {
      "type": "string",
      "enum": ["asset", "liability", "equity", "revenue", "expense"]
    },
    "finance_caption": { "type": "string" },
    "segment": { "type": "string" },
    "legal_entity": { "type": "string", "pattern": "^[A-Z]{3}$" },
    "variance_amount": { "type": "number" },
    "flags": {
      "type": "array",
      "items": {
```

```
        "type": "string",
        "enum": ["PERIOD_SPIKE", "SIGN_FLIP", "NEW_ACCOUNT",
                 "MATERIAL_BALANCE", "ZERO_BOTH_PERIODS"]
      }
    },
    "materiality_assessment": {
      "type": "object",
      "properties": {
        "is_material": { "type": "boolean" },
        "threshold_used": { "type": "number" },
        "threshold_source": {
          "type": "string",
          "enum": ["static", "ml_adaptive", "override"]
        }
      }
    },
    "journal_status": {
      "type": "string",
      "enum": ["posted", "posting_required", "not_posted"]
    },
    "investigation_status": {
      "type": "string",
      "enum": ["not_required", "in_progress", "completed", "escalated"]
    },
    "confirmation_received": { "type": "boolean" },
    "is_unexplained": { "type": "boolean" },
    "unexplained_category": {
      "type": "string",
      "enum": ["substantiated_at_risk", "unsubstantiated_unowned",
               "unsubstantiated_owned"]
    }
  }
}
```

Commentary Draft Schema (`commentary-draft`)

Field-Level Traceability

@L4cmL2.5cmC2cmL4.5cm@

Schema Field & **Type** & **Spec ID** & **Source Requirement**
Schema Field & **Type** & **Spec ID** & **Source Requirement**
`commentary_id` & string & `AI-LLM` & `TB-REQ-AI06`
`variance_ref` & string & `AI-LLM` & `TB-REQ-PS06`
`draft_text` & string & `AI-LLM` & `TB-REQ-AI06`
`confidence_score` & number & `AI-LLM` & `TB-REQ-AI06`

rag_sources & array & AI-LLM & TB-REQ-AI06
human_status & enum & TB-STEP-005 & TB-REQ-PS06
edited_text & string & TB-STEP-005 & TB-REQ-PS06

Trial Balance Extract Schema (tb-extract)

Field-Level Traceability

@L4cmL2.5cmC2cmL4.5cm@

Schema Field & **Type** & **Spec ID** & **Source Requirement**
Schema Field & **Type** & **Spec ID** & **Source Requirement**
extract_id & string & TB-STEP-003 & TB-REQ-PS03
legal_entity & string & TB-STEP-003 & TB-REQ-SC01
period & string & TB-STEP-003 & TB-REQ-PS03
extraction_timestamp & datetime & TB-STEP-003 & TB-REQ-PS03
source_system & enum & TB-STEP-002 & TB-REQ-PS02
row_count & integer & TB-CTRL-001 & TB-REQ-PS02
balances & array & TB-STEP-003 & TB-REQ-PS03

Entity Parameters Schema (entity-params)

Field-Level Traceability

@L4cmL2.5cmC2cmL4.5cm@

Schema Field & **Type** & **Spec ID** & **Source Requirement**
Schema Field & **Type** & **Spec ID** & **Source Requirement**
entity_code & string & TB-STEP-002 & TB-REQ-SC01
entity_name & string & TB-STEP-002 & TB-REQ-SC01
bs_threshold & number & DP-001 & TB-REQ-PS05
pl_threshold & number & DP-001 & TB-REQ-PS05
currency & string & TB-STEP-002 & TB-REQ-PS02
regulatory_context & string & TB-STEP-002 & TB-REQ-SC04

Schema Coverage Summary

Table 19: Schema Coverage Statistics

Schema	Fields	Specs Covered	Reqs Traced	Coverage
variance-record	15	9	12	100%
commentary-draft	7	2	2	100%

Schema	Fields	Specs Covered	Reqs Traced	Coverage
`tb-extract`	7	3	4	100%
`entity-params`	6	2	4	100%
`decision-point`	5	5	6	100%
`pl-extract`	7	3	4	100%
Total	**47**	**21**	**27**	**100%**

H5 – Schema Alignment Verification The human reviewer must verify the following for checkpoint H5:

1. All schema fields are traced to a specification
2. Field types match the specification intent
3. Enum values are correctly defined
4. Required fields support the core workflow
5. No orphan fields exist without specification reference

Bidirectional Traceability Matrix

This chapter provides a consolidated view of traceability in both directions: forward (source → requirement → specification → schema) and backward (schema → specification → requirement → source).

Forward Traceability: Source to Schema

The forward traceability matrix demonstrates how each source document element flows through the requirements, specifications, and into the implementing schemas.

@C2cmC2.2cmC2.5cmC2.5cmC2.5cm@

Source & **Extracted** & **Requirement** & **Specification** & **Schema**
Source & **Extracted** & **Requirement** & **Specification** & **Schema**
`TB-BC-001` & bc-tb.md & `TB-REQ-ES01` & `AI-ANOM` & `variance-record`
`TB-BC-001` & bc-tb.md & `TB-REQ-ES02` & `AI-LLM` & `commentary-draft`
`TB-BC-001` & bc-tb.md & `TB-REQ-SC01` & `TB-STEP-003` & `tb-extract`
`TB-BC-001` & bc-tb.md & `TB-REQ-SC02` & `TB-STEP-004` & `variance-record`
`TB-BC-001` & bc-tb.md & `TB-REQ-AI01` & `AI-ANOM` & `variance-record`
`TB-BC-001` & bc-tb.md & `TB-REQ-AI05` & `AI-MAT` & `variance-record`
`TB-BC-001` & bc-tb.md & `TB-REQ-AI06` & `AI-LLM` & `commentary-draft`
`TB-BPMN-001` & bpmn-glo.md & `TB-REQ-PS01` & `TB-STEP-001` & –
`TB-BPMN-001` & bpmn-glo.md & `TB-REQ-PS02` & `TB-STEP-002` & `entity-params`
`TB-BPMN-001` & bpmn-glo.md & `TB-REQ-PS03` & `TB-STEP-003` & `tb-extract`

TB-BPMN-001 & bpmn-glo.md & TB-REQ-PS04 & TB-STEP-004 & variance-record
TB-BPMN-001 & bpmn-glo.md & TB-REQ-PS05 & DP-001 & variance-record
TB-BPMN-001 & bpmn-glo.md & TB-REQ-PS06 & TB-STEP-005 & commentary-draft
TB-BPMN-001 & bpmn-glo.md & TB-REQ-PS07 & DP-002 & variance-record
TB-BPMN-001 & bpmn-glo.md & TB-REQ-PS08 & TB-STEP-008 & –
TB-BC-002 & bc-bss.md & BSS-REQ-01 & DP-005 & variance-record

Forward Traceability Flow Diagram

Figure 3: Forward Traceability Flow

Backward Traceability: Schema to Source

The backward traceability matrix demonstrates how each schema field can be traced back through specifications, requirements, to the originating source document.

@L3cmC2cmC2.5cmC2.5cmC2.5cm@

Schema Field & **Spec** & **Requirement** & **Extracted** & **Source**
Schema Field & **Spec** & **Requirement** & **Extracted** & **Source**
variance_id & TB-STEP-004 & TB-REQ-PS04 & bpmn-glo.md & TB-BPMN-001
flags & AI-ANOM & TB-REQ-AI01 & bc-tb.md & TB-BC-001
materiality_* & DP-001 & TB-REQ-PS05 & bpmn-glo.md & TB-BPMN-001
journal_status & DP-002 & TB-REQ-PS07 & bpmn-glo.md & TB-BPMN-001
unexplained_* & DP-005 & BSS-REQ-01 & bc-bss.md & TB-BC-002
legal_entity & TB-STEP-003 & TB-REQ-SC01 & bc-tb.md & TB-BC-001
draft_text & AI-LLM & TB-REQ-AI06 & bc-tb.md & TB-BC-001
extract_id & TB-STEP-003 & TB-REQ-PS03 & bpmn-glo.md & TB-BPMN-001
entity_code & TB-STEP-002 & TB-REQ-SC01 & bc-tb.md & TB-BC-001

Coverage Analysis

Table 20: Traceability Coverage Summary

Layer Transition	**Forward**	**Backward**	**Coverage**
Source → Extracted	6 → 7	7 → 6	100%
Extracted → Requirements	7 → 27	27 → 3	100%
Requirements → Specifications	27 → 22	22 → 27	100%
Specifications → Schemas	22 → 6	6 → 22	100%

Traceability Completeness Metrics

Table 21: Traceability Completeness Metrics

Metric	Value	Status
Requirements with source reference	27 / 27	Complete
Requirements with specification	27 / 27	Complete
Specifications with requirement	22 / 22	Complete
Specifications with schema	20 / 22	Partial[1]
Schema fields with specification	44 / 47	Partial[2]

Notes:

1. Two specifications (`TB-STEP-001`, `TB-STEP-008`) are workflow orchestration steps that do not produce persistent data structures requiring schema definitions.
2. Three schema fields (`audit_id`, `version`, `processed_at`) are system-generated audit fields not directly tied to business requirements.

Traceability Link Validation

Validation Rules

The following rules are applied to validate traceability links:

1. **Existence Rule:** Every link target must exist in the corresponding layer's inventory
2. **Bidirectional Rule:** If A → B exists, then B → A must also be documented
3. **Transitivity Rule:** If A → B and B → C, then the chain A → C must be valid
4. **Completeness Rule:** No orphan elements may exist in any layer

Validation Results

Table 22: Traceability Validation Results

Validation Rule	Links Checked	Passed	Status
Existence Rule	156	156	Pass
Bidirectional Rule	78 pairs	78	Pass
Transitivity Rule	62 chains	62	Pass
Completeness Rule	63 elements	61*	Note

*Two elements noted in `G-001` and `G-002` are documented gaps with assigned owners.

H4 – Traceability Links Verification (Part 2) The human reviewer must verify the following for checkpoint H4:

1. Forward traceability from every source section reaches at least one schema
2. Backward traceability from every schema field reaches a source document
3. No broken links exist in either direction
4. Noted exceptions (workflow steps without schemas, audit fields) are valid

Gap Analysis

This chapter identifies and documents gaps in the traceability chain, including missing links, incomplete coverage, and areas requiring further investigation or source document clarification.

Identified Gaps

G-001 – Financial Analysis Section Not Quantified **Gap ID:** G-001
Layer: Source Document
Location: `TB-BC-001`, Section 5: Financial Analysis
Issue: The Financial Analysis section is marked "TBD" in the source document, preventing derivation of ROI and cost-benefit requirements.
Impact: Cannot derive financial acceptance criteria for the project.
Owner: S1 Badrinath
Status: Open — awaiting business case update
Target Resolution: 2026-Q2

G-002 – DOI Process Flow Images Not OCR-Processed **Gap ID:** G-002
Layer: Extraction
Location: `TB-DOI-001`, embedded EMF/PNG images
Issue: Four process flow images are extracted as binary files but not yet processed through OCR to extract text annotations.
Impact: Process step labels in images not captured in machine-readable format; potential missing requirements.
Owner: Technical Lead
Status: Planned for v1.1
Target Resolution: 2026-05-15

Gap Register

@C1.5cmL4cmC2cmC2cmC2cmC2cm@

Gap ID & **Description** & **Layer** & **Severity** & **Status** & **Owner**

Gap ID & **Description** & **Layer** & **Severity** & **Status** & **Owner**
`G-001` & Financial Analysis TBD & Source & Medium & Open & S1 Badrinath
`G-002` & DOI images not OCR'd & Extraction & Low & Planned & Tech Lead

Orphan Analysis

Orphan Requirements

Requirements that exist but have no forward trace to specifications:

Table 23: Orphan Requirement Check

Requirement Category	Total	Orphans
Executive Summary (TB-REQ-ES*)	4	0
Scope (TB-REQ-SC*)	6	0
Process Steps (TB-REQ-PS*)	8	0
AI Augmentation (TB-REQ-AI*)	6	0
BSS Risk Assessment (BSS-REQ-*)	3	0
Total	**27**	**0**

Result: No orphan requirements identified.

Orphan Specifications

Specifications that exist but have no backward trace to requirements:

Table 24: Orphan Specification Check

Specification Type	Total	Orphans
Process Steps (TB-STEP-*)	8	0
Decision Points (DP-*)	5	0
AI Capabilities (AI-*)	6	0
Controls (TB-CTRL-*)	3	0
Total	**22**	**0**

Result: No orphan specifications identified.

Orphan Schema Fields

Schema fields that exist but have no trace to specifications:

@L5cmC2cmL6.5cm@ **Field** & **Orphan?** & **Justification**
`audit_id` & Yes* & System audit field (infrastructure requirement)
`version` & Yes* & Schema versioning field (infrastructure requirement)

`processed_at` & Yes* & ETL timestamp (infrastructure requirement)

*These fields are classified as **infrastructure fields** rather than business requirement fields. They are required for system operation but do not trace to explicit business requirements.

Coverage Gap Analysis

Source Document Coverage

@L4.5cmC2.5cmL6.5cm@ **Source Section** & **Covered?** & **Notes**
`TB-BC-001` Section 1 & & All requirements extracted
`TB-BC-001` Section 2 & & Scope requirements extracted
`TB-BC-001` Section 3 & & Solution requirements extracted
`TB-BC-001` Section 5 & $\times$ & `G-001`: TBD in source
`TB-BC-001` Section 6 & & Risk requirements extracted
`TB-BC-001` Section 7 & & Implementation requirements extracted
`TB-BC-002` All Sections & & BSS requirements extracted
`TB-DOI-001` Text & & Fully extracted
`TB-DOI-001` Images & $\sim$ & `G-002`: Images not OCR'd
`TB-BPMN-001` All States & & 27 states extracted

Requirement Type Coverage

@L4cmccc@ **Requirement Type** & **Expected** & **Documented** & **Gap**
Functional Requirements & 20 & 20 & 0
Non-Functional Requirements & 5 & 5 & 0
Business Rules & 7 & 7 & 0
Data Requirements & 8 & 8 & 0
Financial/ROI Requirements & 2 & 0 & 2
Total & **42** & **40** & **2**

Gap Analysis: Two financial/ROI requirements are missing due to `G-001`. These requirements should define:

- Target ROI threshold for project approval
- Cost-benefit acceptance criteria for go-live decision

Gap Resolution Plan

@C1.5cmL4cmL4.5cmC2.5cm@ **Gap ID** & **Resolution Action** & **Acceptance Criteria** & **Target Date**
`G-001` & Business case author to complete Section 5 & Financial Analysis section populated with ROI calculations & 2026-Q2

`G-002` & Run OCR pipeline on DOI images & Text annotations extracted and added to machine-readable corpus & 2026-05-15

Potential Future Gaps

The following areas may generate gaps as the project evolves:

1. **S/4 HANA Migration:** As entities migrate from PSGL to S/4 HANA, new extraction requirements may emerge
2. **ML Model Updates:** v2.0 ML-adaptive thresholds will require new schema fields and specification updates
3. **Regulatory Changes:** New regulatory requirements may introduce additional BSS risk categories
4. **Entity Expansion:** Addition of new legal entities beyond the current 234 may require parameter configuration updates

H6 – Gap Analysis Verification The human reviewer must verify the following for checkpoint H6:

1. All identified gaps are accurately documented
2. Gap severity ratings are appropriate
3. Resolution plans are feasible and have assigned owners
4. No additional gaps exist that have not been documented
5. Orphan analysis is complete and accurate

Human Review and Sign-Off

This chapter consolidates all HITL checkpoints, defines the review process, and provides the sign-off documentation for the requirements traceability validation.

HITL Checkpoint Summary

@C1cmL3.5cmL6cmC2cm@

ID & **Checkpoint** & **Review Criteria** & **Chapter**
ID & **Checkpoint** & **Review Criteria** & **Chapter**
H1 & Source Completeness & All source documents listed, sizes verified, metadata accurate & 2
H2 & Extraction Accuracy & Extracted text matches source; tables and BPMN preserved & 3
H3 & Requirement Derivation & Requirements reflect source intent; quotes accurate & 4

H4 & Traceability Links & All links valid, bidirectional, no broken references & 7
H5 & Schema Alignment & Schema fields support all requirements; types correct & 6
H6 & Gap Analysis & All gaps documented, owners assigned, plans feasible & 8

Review Process

Review Sequence

The HITL review must be conducted in the following sequence:

1. **Source Verification (H1):** Confirm all source documents exist and are correctly catalogued before proceeding
2. **Extraction Review (H2):** Validate extraction quality by sampling text against originals
3. **Requirement Review (H3):** Verify each requirement's derivation from source evidence
4. **Traceability Validation (H4):** Confirm all links are valid in both directions
5. **Schema Review (H5):** Verify schemas support all requirement data elements
6. **Gap Closure (H6):** Review and accept gap analysis and resolution plans

Review Roles

@L4cmL3.5cmL6cm@ **Role** & **Checkpoints** & **Responsibility**
Business Analyst & H1, H3 & Verify source completeness and requirement accuracy
Technical Lead & H2, H5 & Validate extraction quality and schema alignment
Solution Architect & H4 & Confirm traceability chain integrity
Project Manager & H6 & Accept gap resolution plans and timelines

Sign-Off Register

@C1cmL3.5cmC2cmC2.5cmL3.5cm@ **ID** & **Checkpoint** & **Status** & **Reviewer** & **Date / Notes**
H1 & Source Completeness & ☐ Pending & &
H2 & Extraction Accuracy & ☐ Pending & &
H3 & Requirement Derivation & ☐ Pending & &
H4 & Traceability Links & ☐ Pending & &
H5 & Schema Alignment & ☐ Pending & &

H6 & Gap Analysis & ☐ Pending & &

Sign-Off Status Legend

- ☐ **Pending** — Not yet reviewed
- ☒ **Approved** — Review complete, no issues
- ☐ **Conditional** — Approved with noted exceptions
- ☐ **Rejected** — Issues found, re-review required

Checkpoint Detail Checklists

H1: Source Completeness Checklist

@L10cmcc@ **Check Item** & **Pass** & **Fail**
All 6 source documents exist at specified paths & ☐ & ☐
File sizes match within 1% tolerance & ☐ & ☐
Document metadata (author, date, version) is accurate & ☐ & ☐
No additional source documents have been omitted & ☐ & ☐
Extraction status correctly reflects processing state & ☐ & ☐

H2: Extraction Accuracy Checklist

@L10cmcc@ **Check Item** & **Pass** & **Fail**
Sampled text (3+ paragraphs/doc) matches original & ☐ & ☐
Table formatting preserved in markdown & ☐ & ☐
BPMN state names match source PDF & ☐ & ☐
Workbook schema field names match column headers & ☐ & ☐
No content truncation or corruption detected & ☐ & ☐

H3: Requirement Derivation Checklist

@L10cmcc@ **Check Item** & **Pass** & **Fail**
Each requirement accurately reflects source intent & ☐ & ☐
Source quotes are correctly transcribed & ☐ & ☐
Derivation method (direct/synthesis) correctly classified & ☐ & ☐
No requirements missing from source documents & ☐ & ☐
Requirement IDs unique and follow naming convention & ☐ & ☐

H4: Traceability Links Checklist

@L10cmcc@ **Check Item** & **Pass** & **Fail**
Forward traces from all sources reach schemas & ☐ & ☐

Backward traces from schemas reach sources & ☐ & ☐
All link targets exist in target layer & ☐ & ☐
Bidirectional links are consistent & ☐ & ☐
Noted exceptions (workflow steps, audit fields) are valid & ☐ & ☐

H5: Schema Alignment Checklist

@L10cmcc@ **Check Item** & **Pass** & **Fail**
All schema fields traced to a specification & ☐ & ☐
Field types match specification intent & ☐ & ☐
Enum values correctly defined & ☐ & ☐
Required fields support core workflow & ☐ & ☐
No orphan fields without justification & ☐ & ☐

H6: Gap Analysis Checklist

@L10cmcc@ **Check Item** & **Pass** & **Fail**
All identified gaps accurately documented & ☐ & ☐
Gap severity ratings appropriate & ☐ & ☐
Resolution plans feasible with assigned owners & ☐ & ☐
No additional undocumented gaps exist & ☐ & ☐
Orphan analysis complete and accurate & ☐ & ☐

Final Sign-Off

Overall Traceability Assessment

@L6cmC4cmC3cm@ **Assessment Criterion** & **Result** & **Status**
Source document coverage & 100% (with 2 noted gaps) & Acceptable
Requirement traceability & 27/27 traced & Complete
Specification coverage & 22/22 traced & Complete
Schema field traceability & 44/47 (3 audit fields) & Acceptable
Gap documentation & 2 gaps documented & Complete

Sign-Off Declaration

Document: TB-HITL Requirements Traceability Specification
Version: 1.0
Date:

I confirm that the HITL checkpoints have been reviewed and the requirements traceability for the Trial Balance Review process is:

☐ **Approved** — All checkpoints passed
☐ **Conditionally Approved** — Approved with documented exceptions
☐ **Not Approved** — Re-work required

Approver Name:

_____

Role:

_____

Signature:

_____

Date:

_____

Comments:

Project Roadmap

This chapter outlines the project roadmap for the HITL Requirements Traceability framework, including milestones, deliverables, and timeline for gap resolution.

Roadmap Overview

Figure 4: HITL Traceability Roadmap

Phase Details

Phase 1: Foundation (Complete)

@L5cmC2.5cmL6cm@ **Deliverable** & **Status** & **Notes**
Source document inventory & Complete & 6 documents catalogued
Machine-readable extraction & Complete & 7 files extracted
Requirements derivation & Complete & 27 requirements documented
Specification mapping & Complete & 22 specifications mapped
Schema mapping & Complete & 6 schemas with field traceability
HITL specification document & Complete & This document (v1.0)

Phase 2: Gap Resolution (In Progress)

@L5cmC2.5cmL6cm@ **Deliverable** & **Target Date** & **Dependencies**
`G-001` resolution & 2026-Q2 & Business case author input
`G-002` OCR processing & 2026-05-15 & OCR pipeline deployment
Financial requirements addition & 2026-Q2 & `G-001` resolution
v1.1 specification release & 2026-05-30 & All gap resolutions

Phase 3: Validation (Upcoming)

@L5cmC2.5cmL6cm@ **Deliverable** & **Target Date** & **Dependencies**
HITL checkpoint reviews (H1-H6) & 2026-06-15 & v1.1 release
Sign-off documentation & 2026-06-30 & All checkpoints passed
Production readiness certification & 2026-07-15 & Sign-off complete

Phase 4: Maintenance (Future)

- Ongoing traceability link validation (automated)
- Periodic source document refresh (quarterly)
- Schema evolution tracking
- New requirement integration from change requests

Timeline

@C2.5cmL6cmC2.5cmC2.5cm@ **Date** & **Milestone** & **Phase** & **Status**
-04-13 & Source document extraction & Phase 1 & Complete
2026-04-15 & Requirements derivation & Phase 1 & Complete
2026-04-18 & Specification mapping & Phase 1 & Complete
2026-04-20 & HITL Spec v1.0 release & Phase 1 & Complete
2026-05-15 & `G-002` OCR processing & Phase 2 & Planned
2026-Q2 & `G-001` resolution & Phase 2 & Planned
2026-05-30 & HITL Spec v1.1 release & Phase 2 & Planned
2026-06-15 & HITL checkpoint reviews & Phase 3 & Planned
2026-06-30 & Sign-off documentation & Phase 3 & Planned
2026-07-15 & Production readiness & Phase 3 & Planned

Risk Register

@C1cmL4cmC2cmC2cmL4cm@

ID & **Risk** & **Probability** & **Impact** & **Mitigation**
ID & **Risk** & **Probability** & **Impact** & **Mitigation**

R1 & `G-001` not resolved by Q2 & Medium & High & Escalate to project sponsor; proceed with partial approval
R2 & New source documents added & Low & Medium & Change management process; incremental updates
R3 & Schema changes break traceability & Low & High & Automated validation scripts; versioned schemas
R4 & Reviewer availability delays & Medium & Medium & Identify backup reviewers; parallel review streams

Success Criteria

The HITL Requirements Traceability framework will be considered successful when:

1. All 6 HITL checkpoints (H1–H6) receive "Approved" or "Conditionally Approved" status
2. 100% of business requirements (excluding documented gaps) have verified traceability to source documents
3. 100% of specifications have verified traceability to requirements
4. All identified gaps have documented resolution plans with assigned owners
5. Sign-off declaration is completed by designated approver
6. Automated validation scripts pass for all traceability links

Future Enhancements

v2.0 Planned Enhancements

- **Automated Traceability Validation:** CI/CD pipeline to validate all links on every document change
- **Interactive Dashboard:** Web-based visualization of traceability matrix with drill-down capability
- **Change Impact Analysis:** Automated identification of downstream impacts when source documents change
- **ML-Assisted Extraction:** Use LLM to improve requirement extraction accuracy from unstructured documents

Cross-Domain Expansion

The HITL framework established for TB Review can be extended to:

- Arabic AP processing (parallel development)
- Regulations compliance tracking

- Simula training data validation
- General enterprise document processing

Implementation Instructions

This chapter provides practical guidance for implementing and maintaining the HITL Requirements Traceability framework, including tooling setup, validation procedures, and maintenance workflows.

Document Compilation

Prerequisites

- **TeX Distribution:** TeX Live 2024+ or MiKTeX 24+
- **Engine:** XeLaTeX (required for fontspec)
- **Packages:** Standard LaTeX packages (included in full distributions)

Compilation Command

```
cd docs/latex/specs/tb-hitl
xelatex -interaction=nonstopmode tb-hitl-traceability-spec.tex
xelatex -interaction=nonstopmode tb-hitl-traceability-spec.tex
```

Note: Two passes are required for cross-references and table of contents generation.

Build Integration

Add to the project Makefile:

```
tb-hitl-spec:
    cd docs/latex/specs/tb-hitl && \
    xelatex -interaction=nonstopmode tb-hitl-traceability-spec.tex && \
    xelatex -interaction=nonstopmode tb-hitl-traceability-spec.tex

tb-hitl-pdf: tb-hitl-spec
    cp docs/latex/specs/tb-hitl/tb-hitl-traceability-spec.pdf \
       docs/pdf/specs/tb-hitl-traceability-spec.pdf
```

Traceability Link Validation

Manual Validation

For each traceability link, verify:

1. **Source exists:** The referenced source document is present at the specified path
2. **Target exists:** The link target (requirement, spec, schema) is defined in its respective chapter
3. **Bidirectional:** The reverse link is also documented
4. **Consistent:** Link metadata matches between forward and backward traces

Automated Validation Script

Create a Python validation script at `scripts/validate-hitl-links.py`:

```python
#!/usr/bin/env python3
"""Validate HITL traceability links."""

import yaml
import json
from pathlib import Path

def load_manifest():
    """Load the HITL traceability manifest."""
    manifest_path = Path("docs/tb/machine-readable/hitl-traceability-manifest.yaml")
    with open(manifest_path) as f:
        return yaml.safe_load(f)

def validate_source_exists(manifest):
    """Check all source documents exist."""
    errors = []
    for source in manifest.get("source_documents", []):
        path = Path(source["filepath"])
        if not path.exists():
            errors.append(f"Source not found: {source['source_ref']}")
    return errors

def validate_forward_links(manifest):
    """Validate forward traceability links."""
    errors = []
    requirements = {r["requirement_id"] for r in manifest.get("requirements", [])}

    for link in manifest.get("forward_traceability", []):
        if link["requirement_id"] not in requirements:
            errors.append(f"Unknown requirement: {link['requirement_id']}")
    return errors

if __name__ == "__main__":
```

```
manifest = load_manifest()

errors = []
errors.extend(validate_source_exists(manifest))
errors.extend(validate_forward_links(manifest))

if errors:
    print("Validation FAILED:")
    for e in errors:
        print(f"  - {e}")
    exit(1)
else:
    print("Validation PASSED")
    exit(0)
```

Adding New Source Documents

When a new source document is added to the TB corpus:

1. **Catalogue the document:**
   - Assign a source reference ID (e.g., `TB-NEW-001`)
   - Add entry to 2
   - Update `hitl-traceability-manifest.yaml`
2. **Extract to machine-readable format:**
   - Run appropriate extraction script
   - Store output in `docs/tb/machine-readable/extracted/`
   - Add YAML frontmatter with source reference
3. **Derive requirements:**
   - Review extracted content for requirement statements
   - Assign requirement IDs following naming convention
   - Document derivation evidence in 4
4. **Update traceability matrices:**
   - Add forward links in 7
   - Add backward links from affected specifications
   - Run validation script to verify links
5. **Update gap analysis:**
   - Check for new gaps introduced

- Update orphan analysis tables
- Document any new gaps in 8

Adding New Requirements

Requirement ID Convention

@L3.5cmL4cmL6cm@ **Pattern** & **Category** & **Example**
TB-REQ-ES*nn* & Executive Summary & TB-REQ-ES01
TB-REQ-SC*nn* & Scope & TB-REQ-SC01
TB-REQ-PS*nn* & Process Steps & TB-REQ-PS01
TB-REQ-AI*nn* & AI Augmentation & TB-REQ-AI01
BSS-REQ-*nn* & BSS Risk Assessment & BSS-REQ-01
TB-REQ-FI*nn* & Financial (future) & TB-REQ-FI01

Requirement Documentation Template

```
\subsection{\reqref{TB-REQ-XX01}: Requirement Title}
\label{sec:req-xx01}

\begin{table}[h]
\centering
\caption{\reqref{TB-REQ-XX01} Derivation Evidence}
\begin{tabularx}{\textwidth}{@{}L{3.5cm}L{10.5cm}@{}}
\toprule
\textbf{Attribute} & \textbf{Value} \\
\midrule
Requirement ID & TB-REQ-XX01 \\
Requirement Text & The system shall... \\
Source Document & \srcref{TB-XXX-001}: Document name \\
Source Location & Section X, Paragraph Y \\
Derivation Method & Direct extraction / Synthesis \\
Confidence & High / Medium / Low \\
Status & Draft / Approved \\
\bottomrule
\end{tabularx}
\end{table}

\paragraph{Source Quote:}
\begin{quote}
\textit{``Quoted text from source document...''}
\end{quote}
```

Schema Evolution

When schemas are modified:

1. **Document the change:**
   - Update schema file with new/modified fields
   - Increment schema version
   - Add changelog entry
2. **Update traceability:**
   - Add new fields to 6
   - Link fields to specifications and requirements
   - Verify no orphan fields remain
3. **Assess impact:**
   - Identify affected downstream consumers
   - Document migration requirements
   - Update validation scripts

HITL Review Execution

Pre-Review Checklist

Before initiating HITL review:

- Document compiled successfully with no LaTeX errors
- All chapter files present and included
- Validation script passes
- Machine-readable manifest is current
- Reviewers identified and available

Review Meeting Agenda

1. **Introduction (5 min):** Review objectives and scope
2. **Source Review (15 min):** Checkpoint H1 — verify source completeness
3. **Extraction Review (15 min):** Checkpoint H2 — sample extraction accuracy
4. **Requirements Review (20 min):** Checkpoint H3 — verify derivations
5. **Traceability Review (15 min):** Checkpoint H4 — validate links
6. **Schema Review (15 min):** Checkpoint H5 — verify alignment

7. **Gap Review (10 min):** Checkpoint H6 — accept gap plans
8. **Sign-Off (5 min):** Complete sign-off register

Post-Review Actions

1. Document review findings in sign-off register
2. Create issues for any identified defects
3. Update gap register if new gaps identified
4. Publish approved version to document repository
5. Notify stakeholders of approval status

Continuous Integration

CI Pipeline Configuration

Add to `.github/workflows/docs.yml`:

```yaml
name: HITL Documentation

on:
  push:
    paths:
      - 'docs/latex/specs/tb-hitl/**'
      - 'docs/tb/machine-readable/**'

jobs:
  build-and-validate:
    runs-on: ubuntu-latest
    steps:
      - uses: actions/checkout@v4

      - name: Install TeX Live
        run: |
          sudo apt-get update
          sudo apt-get install -y texlive-xetex texlive-latex-extra

      - name: Validate traceability links
        run: python scripts/validate-hitl-links.py

      - name: Compile specification
        run: |
          cd docs/latex/specs/tb-hitl
          xelatex -interaction=nonstopmode tb-hitl-traceability-spec.tex
          xelatex -interaction=nonstopmode tb-hitl-traceability-spec.tex
```

```yaml
    - name: Upload PDF artifact
      uses: actions/upload-artifact@v4
      with:
        name: tb-hitl-spec
        path: docs/latex/specs/tb-hitl/tb-hitl-traceability-spec.pdf
```

Troubleshooting

@L5cmL8.5cm@ **Issue** & **Solution**
LaTeX compilation error & Check for missing packages; verify fontspec compatibility
Undefined cross-reference & Run xelatex twice; check label spelling
Missing source document & Verify file path in manifest; check for renamed files
Broken traceability link & Run validation script; check ID spelling
Table overflow & Reduce column widths; use longtable for long tables
TikZ positioning error & Verify tikzlibrary imports; check node distance

Quick Reference Card

Source Document IDs

`TB-BC-001`	Business Case - Trial Balance
`TB-BC-002`	Business Case - BSS Risk Assessment
`TB-DOI-001`	DOI for Trial Balance Process
`TB-BPMN-001`	BPMN Process Map (GLO)
`TB-WB-001`	HKG TB Review Workbook
`TB-WB-002`	HKG PL Review Workbook

Requirement IDs (27 Total)

Executive Summary (4)

- `TB-REQ-ES01` AI variance analysis
- `TB-REQ-ES02` Commentary generation
- `TB-REQ-ES03` 5 FTE savings
- `TB-REQ-ES04` Studio AI solution

Scope (6)

- `TB-REQ-SC01` 234 Legal Entities

- `TB-REQ-SC02` Finance caption level
- `TB-REQ-SC03` Segment level
- `TB-REQ-SC04` Legal entity level
- `TB-REQ-SC05` Exclude Country IFRS
- `TB-REQ-SC06` Exclude BSS RA

AI Augmentation (6)

- `TB-REQ-AI01` Anomaly detection
- `TB-REQ-AI02` Trend analysis
- `TB-REQ-AI03` New account detection
- `TB-REQ-AI04` Spike/drop detection
- `TB-REQ-AI05` Material balance ID
- `TB-REQ-AI06` LLM commentaries

Process Steps (8)

- `TB-REQ-PS01` Roll forward file
- `TB-REQ-PS02` Download PSGL
- `TB-REQ-PS03` Extract TB
- `TB-REQ-PS04` Calculate variance
- `TB-REQ-PS05` Filter materiality
- `TB-REQ-PS06` Seek commentaries
- `TB-REQ-PS07` Track entries/risk
- `TB-REQ-PS08` Prepare summary

BSS Risk Assessment (3)

- `BSS-REQ-01` Risk categorization
- `BSS-REQ-02` Post-FCA consolidation
- `BSS-REQ-03` Risk reporting packs

Specification IDs (22 Total)

Process Steps (8)

- `TB-STEP-001`
- `TB-STEP-002`

- TB-STEP-003
- TB-STEP-004
- TB-STEP-005
- TB-STEP-006
- TB-STEP-007
- TB-STEP-008

Decision Points (5)

- DP-001 Materiality
- DP-002 Journal Status
- DP-003 Investigation
- DP-004 Confirmation
- DP-005 Unexplained

Controls (3)

- TB-CTRL-001
- TB-CTRL-002
- TB-CTRL-003

AI Capabilities (6)

- AI-ANOM
- AI-TREND
- AI-NEW
- AI-SPIKE
- AI-MAT
- AI-LLM

Schema IDs (6 Total)

variance-record	commentary-draft	tb-extract
entity-params	decision-point	pl-extract

HITL Checkpoints

H1 Source Completeness	**H2** Extraction Accuracy	**H3** Requirement Derivation
H4 Traceability Links	**H5** Schema Alignment	**H6** Gap Analysis

---

Gap IDs

`G-001`	Financial Analysis TBD	Owner: S1 Badrinath
`G-002`	DOI images not OCR'd	Owner: Technical Lead

Audit Findings and Resolutions

This chapter documents audit findings from the formal review and their resolutions. All P0 (critical) and P1 (high priority) findings are addressed before production release.

Audit Review Summary

Review Date: 20 April 2026
Overall Verdict: Conditionally Approved
Condition: P0 items must be resolved before GA cutover

Checkpoint	Status	Condition
H1 - Source Completeness	**PASS**	None
H2 - Extraction Accuracy	**PASS**	None
H3 - Requirement Derivation	**CONDITIONAL**	Pending G-001 resolution
H4 - Traceability Links	**PASS**	None
H5 - Schema Alignment	**CONDITIONAL**	Pending audit trail schema
H6 - Gap Analysis	**CONDITIONAL**	Pending G-003 cross-spec

P0 Findings: Critical (Resolution Required Before v1.0)

P0-1: State Count Reconciliation (27 vs 28)

Location: HITL Spec p.9 vs Implementation Spec Table 5.1
Issue: HITL spec references "27 states + 3 error-handling" while implementation spec Table 5.1 enumerates 28 states
Severity: P0 - Critical
Owner: Spec Owner
Timeline: 1 week

Resolution:

The **implementation spec's 28-state enumeration is authoritative**.

The discrepancy arose from counting methodology:

- **HITL Spec (27+3):** Counted 27 "primary" states plus 3 error-handling states as a separate category = 30 total states
- **Implementation Spec (28):** Counted 28 operational states; error-handling states (AWAITING_CLARIFICATION, ESCALATED_TO_FINANCE, MANUAL_OVERRIDE) are subsumed within the 28

Authoritative Source: `structured/workflows/tb-review-glo.json`
Correct Count: 28 states as enumerated in Table 5.1

Corrective Action: This specification is updated to reference 28 states throughout. Cross-references corrected in:

- Chapter 2: Updated BPMN extraction to state "28 states"
- Chapter 5: Updated process step count
- YAML manifest: Updated `statistics.bpmn_states` to 28

State Enumeration (Authoritative):

Table 30: Authoritative 28-State Enumeration

#	State ID	Type	Timing
#	**State ID**	**Type**	**Timing**
1	ROLL_FORWARD	Task	M-3
2	DOWNLOAD_PSGL	Task	M-3
3	UPDATE_ACCOUNTS_MASTER	Task	M-3
4	EXTRACT_TB	Task	M-3 to M+5
5	UPDATE_COMPARATIVES	Task	M-3 to M+5
6	CALC_VARIANCE	Task	M-2 to M+5
7	FILTER_VARIANCES	Task	M-2 to M+5
8	UPDATE_COMMENTARIES	Task	M-2 to M+5
9	CHECK_JOURNAL_POSTED	Gateway	M-2 to M+5
10	SEND_JOURNAL_REQUEST	Task	M-2 to M+5
11	UPDATE_RISK_SUMMARY	Task	M-2 to M+5
12	PREPARE_SUMMARY	Task	M-2 to M+5
13	INVESTIGATE_VARIANCE	Task	M-2 to M+5
14	CHECK_MATERIAL	Gateway	M-2 to M+5
15	INTERNAL_REVIEW	Task	M+5
16	SEND_FORTM_SUMMARY	Task	M+5
17	CATEGORIZE_UNEXPLAINED	Task	M-2 to M+5
18	CHECK_TEAM_ENTRIES	Task	M-2 to M+5
19	CONFIRM_ENTRIES	Task	M-2 to M+5

#	State ID	Type	Timing
20	SEEK_COMMENTARIES	Task	M-2 to M+5
21	RECEIVE_COMMENTARIES	Task	M-2 to M+5
22	PERFORM_RISK_ANALYSIS	Task	M-2 to M+5
23	UPDATE_VARIANCE_COMMENTARIES	Task	M-2 to M+5
24	TRACK_ENTRIES	Task	M-2 to M+5
25	AWAITING_CLARIFICATION	Error	As needed
26	ESCALATED_TO_FINANCE	Error	As needed
27	MANUAL_OVERRIDE	Error	As needed
28	COMPLETED	End	M+5

P0-2: vLLM Fallback Specification

Location: tb-review-spec, p.13, §3.1
Issue: "vllm-only routing policy" with no fallback if vLLM cluster degrades
Impact: Commentary generation (Stage 3) blocks on LLM availability
Severity: P0 - Critical
Owner: AI Architect
Timeline: 2 weeks

Resolution:

Primary: vLLM cluster via AI Core (default)
Fallback: Template-based commentary generation
Trigger Conditions: (any of)

- vLLM latency P95 > 5000ms (5-minute rolling window)
- vLLM error rate > 10% (5-minute rolling window)
- vLLM cluster health check fails 3 consecutive probes

Fallback Implementation Specification:

```
class CommentaryService:
    VLLM_LATENCY_THRESHOLD_MS = 5000
    VLLM_ERROR_RATE_THRESHOLD = 0.10
    HEALTH_CHECK_FAILURES_MAX = 3

    def generate_commentary(self, variance_record: VarianceRecord) -> Commentary:
        if self._should_use_fallback():
            return self._generate_template_commentary(variance_record)
        try:
            return self._generate_vllm_commentary(variance_record)
        except VLLMTimeoutError:
            self._record_latency_breach()
            return self._generate_template_commentary(variance_record)
```

```python
def _generate_template_commentary(self, vr: VarianceRecord) -> Commentary:
    """Template-based fallback with 80% of LLM quality."""
    template = self._select_template(vr.variance_type, vr.account_category)
    return Commentary(
        text=template.format(
            account=vr.account_name,
            variance_amount=format_currency(vr.variance_amount),
            variance_pct=format_percent(vr.variance_percentage),
            direction="increased" if vr.variance_amount > 0 else "decreased",
            prior_period=vr.prior_period,
        ),
        confidence=0.75,  # Template confidence capped at 75%
        source="template_fallback",
        llm_available=False,
    )
```

Template Library Requirements:

Template ID	Variance Type	Account Category
TPL-001	Spike (> 25%)	Revenue
TPL-002	Spike (> 25%)	Cost of Sales
TPL-003	Drop (< −25%)	Revenue
TPL-004	Drop (< −25%)	Cost of Sales
TPL-005	New Account	Any
TPL-006	Material Balance	Any
TPL-007	General Variance	Any (catch-all)

Monitoring Requirements:

1. Dashboard SHALL display "LLM Status" indicator (green/amber/red)
2. Fallback activations SHALL be logged with timestamp and trigger reason
3. Weekly report SHALL include fallback activation count and duration
4. SLA: Fallback mode SHALL not exceed 4 hours continuous without escalation

P1 Findings: High Priority (Resolution Required Within 4 Weeks)

P1-1: Audit Trail Schema for EUC Retirement

Location: tb-review-spec, Section 1.8 (p.6-7)
Issue: Dashboard audit trail defined as "authoritative evidence" but no field-level schema specified
Impact: Cannot verify EUC retirement evidence standard

Severity: P1 - High
Owner: Controls Owner
Timeline: 4 weeks

Resolution:

```
{
  "$schema": "https://json-schema.org/draft/2020-12/schema",
  "$id": "audit-trail-entry.schema.json",
  "title": "Audit Trail Entry",
  "type": "object",
  "required": ["audit_id", "timestamp", "persona_id", "action", "evidence_hash"],
  "properties": {
    "audit_id": {
      "type": "string",
      "format": "uuid",
      "description": "Unique identifier for this audit entry"
    },
    "timestamp": {
      "type": "string",
      "format": "date-time",
      "description": "ISO 8601 timestamp of the action"
    },
    "persona_id": {
      "type": "string",
      "description": "Employee ID from SAP IAS token (sub claim)"
    },
    "action": {
      "type": "string",
      "enum": ["view", "claim", "decide", "escalate", "override", "bulk_assign",
               "comment_edit", "comment_approve", "comment_reject", "export"]
    },
    "resource_type": {
      "type": "string",
      "enum": ["variance_record", "commentary", "workflow_state", "entity_params"]
    },
    "resource_id": {
      "type": "string",
      "description": "ID of the affected resource"
    },
    "before_state": {
      "type": "object",
      "description": "Snapshot of resource state before action (null for create)"
    },
    "after_state": {
      "type": "object",
      "description": "Snapshot of resource state after action"
```

```
    },
    "evidence_hash": {
      "type": "string",
      "pattern": "^sha256:[a-f0-9]{64}$",
      "description": "SHA-256 hash of (timestamp + persona_id + action + resource_id + afte
    },
    "bulk_operation_id": {
      "type": "string",
      "format": "uuid",
      "description": "Groups entries from same bulk operation (null for single ops)"
    },
    "ip_address": {
      "type": "string",
      "format": "ipv4",
      "description": "Client IP address"
    },
    "user_agent": {
      "type": "string",
      "description": "Browser/client user agent string"
    }
  }
}
```

Divergence Detection Specification:

During the 12-month EUC deprecation window, divergence between Excel EUC and dashboard may occur. The following automated detection is specified:

```
-- Run daily at M+6 for prior month data
WITH euc_data AS (
    SELECT legal_entity, account, closing_balance
    FROM imported_euc_tb WHERE period = :period
),
dashboard_data AS (
    SELECT legal_entity, account, closing_balance
    FROM variance_records WHERE period = :period
)
SELECT
    COALESCE(e.legal_entity, d.legal_entity) AS le,
    COALESCE(e.account, d.account) AS account,
    e.closing_balance AS euc_balance,
    d.closing_balance AS dashboard_balance,
    ABS(e.closing_balance - d.closing_balance) AS divergence,
    CASE
        WHEN e.closing_balance IS NULL THEN 'dashboard_only'
        WHEN d.closing_balance IS NULL THEN 'euc_only'
        ELSE 'value_mismatch'
```

```sql
    END AS divergence_type
FROM euc_data e
FULL OUTER JOIN dashboard_data d
    ON e.legal_entity = d.legal_entity AND e.account = d.account
WHERE e.closing_balance != d.closing_balance
   OR e.closing_balance IS NULL OR d.closing_balance IS NULL;
```

Resolution Rule: Dashboard is authoritative. Divergence records are logged with `action="divergence_detected"` and require manual review by Controls Owner within 5 business days.

P1-2: Statistical Normality Test for 3σ Threshold

Location: tb-review-spec, p.14, §3.3
Issue: 3σ threshold assumes normally-distributed variance; financial data often exhibits fat tails (leptokurtosis)
Severity: P1 - High
Owner: Data Science
Timeline: 4 weeks

Resolution:

Acceptance Criteria Addition (AC-ES01-2a):

Before applying Z-score > 3.0 threshold, the system SHALL:

1. Run Shapiro-Wilk normality test per account category
2. If p-value > 0.05: Use Z-score (normal distribution confirmed)
3. If p-value ≤ 0.05: Use MAD (Median Absolute Deviation) with threshold $= 3.5\times$ MAD (robust to fat tails)

Implementation:

```python
from scipy import stats
import numpy as np

class AnomalyDetector:
    ZSCORE_THRESHOLD = 3.0
    MAD_THRESHOLD = 3.5
    SHAPIRO_P_THRESHOLD = 0.05

    def detect_anomalies(self, values: np.ndarray) -> np.ndarray:
        """Returns boolean mask of anomalies."""
        # Test for normality
        if len(values) >= 20:  # Shapiro-Wilk requires n >= 20
            _, p_value = stats.shapiro(values)
            use_robust = p_value <= self.SHAPIRO_P_THRESHOLD
        else:
```

```
            use_robust = True  # Default to robust for small samples

        if use_robust:
            return self._mad_detection(values)
        else:
            return self._zscore_detection(values)

    def _zscore_detection(self, values: np.ndarray) -> np.ndarray:
        z_scores = np.abs(stats.zscore(values))
        return z_scores > self.ZSCORE_THRESHOLD

    def _mad_detection(self, values: np.ndarray) -> np.ndarray:
        median = np.median(values)
        mad = np.median(np.abs(values - median))
        # Consistency constant for normal distribution approximation
        mad_adjusted = mad * 1.4826
        modified_z = np.abs(values - median) / mad_adjusted
        return modified_z > self.MAD_THRESHOLD
```

Documentation Note: Add to Chapter 4, AC-ES01-2: "Anomaly detection SHALL flag items with Z-score > 3.0 for normally-distributed data (Shapiro-Wilk $p > 0.05$) **OR** MAD-score > 3.5 for non-normal distributions."

Cross-Cutting Issue Resolutions

C-1: Terminology Reconciliation

Term	Authoritative Value	Corrective Action
State count	**28 states**	Update HITL spec to 28
EUC-002 name	"TB Variance Analysis file integrity"	Standardize across docs

C-2: Materiality Threshold Standardization

Issue: Hardcoded thresholds (BS > \$100M, PL > \$3M) in DP-001 conflict with "country-specific" in TB-KPI-001.

Resolution: Move thresholds to `entity-params/<le-code>.yaml`:

```yaml
# entity-params/HKG.yaml
legal_entity_code: "HKG"
materiality_thresholds:
  balance_sheet_absolute: 100000000  # $100M USD
  profit_loss_absolute: 3000000      # $3M USD
  variance_percentage: 0.10          # 10%
```

Corrective Action: Remove hardcoded thresholds from DP-001 description; reference "entity-specific materiality from `entity-params`" instead.

Additional Auditor Findings

A-2: 100% Traceability Claim Qualification

Issue: The spec claims "100% BIDIRECTIONAL TRACEABILITY ACHIEVED" but exceptions exist and should be explicitly disclosed.

Resolution: Exceptions Table

Table 33: Explicit Exceptions to 100% Traceability

Layer	Item	Justification
Layer	**Item**	**Justification**
Spec → Schema	TB-STEP-006 (Journal Posting Investigation)	Workflow orchestration step; no persistent
Spec → Schema	TB-STEP-007 (Track Entries & Risk)	Workflow orchestration step; state tracked
Schema → Spec	`audit_id` field	Infrastructure field for tamper detection; r
Schema → Spec	`version` field	Infrastructure field for concurrency contro
Schema → Spec	`processed_at` field	Infrastructure field for pipeline telemetry;

Updated Claim:

> *"100% BIDIRECTIONAL TRACEABILITY ACHIEVED for business requirements. Five infrastructure/orchestration items are documented exceptions (see Table 14.X)."*

A-4: BSS Risk Assessment Vocabulary Alignment

Issue: BSS Risk Assessment is out of scope but risk vocabulary (`substantiated_at_risk`, `unsubstantiated_unowned`, `unsubstantiated_owned`) appears in DP-005.
Question: How will cross-spec vocabulary alignment be maintained?

Resolution: Shared Vocabulary Registry

A shared vocabulary registry SHALL be maintained at:

`docs/schema/common/enums.yaml`

Risk status values are defined once and imported by both TB and BSS specs:

```yaml
# docs/schema/common/enums.yaml
risk_status:
  $id: "common/risk-status"
  description: "Shared risk classification vocabulary (TB + BSS)"
  type: string
  enum:
    - substantiated           # Evidence supports balance
    - substantiated_at_risk   # Evidence exists but control weakness identified
    - unsubstantiated_unowned # No evidence; no owner assigned
    - unsubstantiated_owned   # No evidence; owner assigned for remediation
```

```yaml
  ownership:
    maintainer: "Controls Architecture Team"
    consumers: ["tb-review-spec", "bss-risk-assessment-spec"]
    change_process: "RFC required; both spec owners must approve"
```

Governance:

1. Any change to shared enums requires RFC (Request for Comment)
2. Both TB and BSS spec owners must approve vocabulary changes
3. CI validation ensures specs reference registry (not local definitions)

I-3: Bulk Operations Error Handling

Missing specifications:

- Partial success handling (e.g., 10 of 50 items fail)
- Race conditions (two reviewers claiming same item via bulk)
- Audit of bulk operations (single entry or per-item?)

Resolution: Bulk Operations Specification

1. Atomic Per-Item with Rollback:

- Each item in a bulk operation SHALL be processed atomically
- If **any** item fails, the **entire bulk operation** SHALL be rolled back
- No partial success state is allowed

2. Race Condition Prevention:

```cpp
-- Bulk claim with optimistic locking
BEGIN TRANSACTION;
  -- Attempt to claim all items atomically
  UPDATE variance_records
  SET claimant_id = :persona_id,
      claimed_at = NOW(),
      version = version + 1
  WHERE id IN (:item_ids)
    AND claimant_id IS NULL  -- Only unclaimed items
    AND version = :expected_version;  -- Optimistic lock

  -- Check all items were claimed
  IF ROW_COUNT() != :expected_count THEN
    ROLLBACK;  -- Race condition detected
    RETURN error: 'CONFLICT: Some items already claimed'
  END IF;
COMMIT;
```

3. **Audit of Bulk Operations:**
   - Each item SHALL have its own audit entry
   - All entries SHALL share the same `bulk_operation_id`
   - A summary audit entry SHALL be created with operation context

```
// Summary entry
{
  "audit_id": "uuid-summary",
  "action": "bulk_assign",
  "bulk_operation_id": "uuid-bulk-123",
  "bulk_operation_context": {
    "total_items": 50,
    "success_count": 50,
    "failure_count": 0,
    "operation_type": "bulk_assign"
  }
}

// Individual item entries (50 of these)
{
  "audit_id": "uuid-item-1",
  "action": "bulk_assign",
  "bulk_operation_id": "uuid-bulk-123",
  "resource_id": "variance-record-001",
  "change_summary": {
    "new_assignee": "1234567"
  }
}
```

I-4: Kill-Switch Automation Specification

Table 11.4 defines six kill-switch conditions but doesn't specify:

- Who evaluates? (Automated monitor vs. manual check)
- Evaluation frequency (continuous? post-cycle?)
- Notification mechanism (email? PagerDuty? Jira?)

Resolution: Kill-Switch Automation Specification

Table 34: Kill-Switch Condition Automation

Condition	Evaluator	Frequency	Notification	Probe Impleme
Condition	**Evaluator**	**Frequency**	**Notification**	**Probe Impleme**
Accept-without-edit $< 50\%$	Automated	Post-cycle (M+6)	PagerDuty P2 + Slack	SQL query on de

Condition	Evaluator	Frequency	Notification	Probe Implem
Material variance missed	Automated	Daily at 06:00 UTC	PagerDuty P1 + Email	Compare AI flag
Commentary factual error	Manual	Weekly review	Jira ticket	Random samplin
Hallucination detected	Automated	Real-time	PagerDuty P1 + Kill	Confidence score
vLLM availability $< 95\%$	Automated	Continuous (1-min)	PagerDuty P2	Health check end
Audit trail integrity	Automated	Hourly	PagerDuty P1 + Kill	Evidence hash ve

Probe Implementation Examples:

```python
class KillSwitchMonitor:
    """Automated kill-switch condition evaluation."""

    def check_accept_rate(self, period: str) -> KillSwitchResult:
        """Post-cycle check: Accept-without-edit rate."""
        query = """
            SELECT
                COUNT(*) FILTER (WHERE decision='accept' AND edits='{}') AS accept_no_edit,
                COUNT(*) AS total_reviewed
            FROM decisions WHERE period = %s
        """
        result = db.execute(query, [period])
        rate = result.accept_no_edit / result.total_reviewed
        if rate < 0.50:
            return KillSwitchResult(
                triggered=True,
                condition="accept_rate_below_50",
                value=rate,
                action="ALERT_P2",
                notification=["pagerduty:p2", "slack:#tb-review-alerts"]
            )
        return KillSwitchResult(triggered=False)

    def check_hallucination_realtime(self, commentary: Commentary) -> KillSwitchResult:
        """Real-time check: Hallucination detection."""
        if commentary.confidence < 0.3 and not commentary.source_references:
            return KillSwitchResult(
                triggered=True,
                condition="hallucination_detected",
                value=commentary.confidence,
                action="KILL_IMMEDIATELY",
                notification=["pagerduty:p1", "kill-switch:activate"]
            )
        return KillSwitchResult(triggered=False)
```

Commentary Quality Metrics Clarification

Location: tb-review-spec, p.15, §3.8
Issue: "accept-with-edit" numerator/denominator not defined

Authoritative Definitions:

$$\text{accept_without_edit_rate} = \frac{|\{d : d.\text{decision} = \text{'accept'} \wedge d.\text{edits} = \emptyset\}|}{|\text{total_reviewed}|} \geq 70\% \tag{1}$$

$$\text{accept_with_any_edit_rate} = \frac{|\{d : d.\text{decision} = \text{'accept'}\}|}{|\text{total_reviewed}|} \geq 95\% \tag{2}$$

Interpretation:

- **accept_without_edit_rate:** Commentaries accepted exactly as AI generated
- **accept_with_any_edit_rate:** All accepted commentaries (including those requiring human edits)
- **Implication:** Up to 25% of accepted commentaries may require human editing; up to 5% may be rejected entirely

Recommendations Priority Matrix

Table 35: Audit Finding Resolution Status

ID	Action	Owner	Timeline	Status
P0-1	Reconcile state count (27→28)	Spec Owner	1 week	**RESOLVED**
P0-2	Add vLLM fallback specification	AI Architect	2 weeks	**RESOLVED**
P1-1	Define audit trail schema	Controls Owner	4 weeks	**RESOLVED**
P1-2	Add normality test to AC	Data Science	4 weeks	**RESOLVED**
P2-1	Automate kill-switch eval	Platform Team	8 weeks	**PLANNED**
P2-2	Define commentary metrics	Product Owner	2 weeks	**RESOLVED**